

MAPPING THE SEMICONDUCTOR VALUE CHAIN

WORKING TOWARDS
IDENTIFYING DEPENDENCIES
AND VULNERABILITIES

OECD SCIENCE, TECHNOLOGY
AND INDUSTRY
POLICY PAPERS

June 2025 **No. 182**

OECD Science, Technology and Industry Policy Papers

This paper was approved and declassified by written procedure by the Digital Policy Committee (DPC) on 12 December 2024 and by the Committee on Industry, Innovation and Entrepreneurship (CIIE) on 17 December 2024. It was prepared for publication by the OECD Secretariat.

Note to Delegations:

This document is also available on O.N.E under the reference code:

DSTI/DPC/CIIE(2024)3/FINAL

This document, as well as any data and map included herein, are without prejudice to the status of or sovereignty over any territory, to the delimitation of international frontiers and boundaries and to the name of any territory, city or area.

The statistical data for Israel are supplied by and under the responsibility of the relevant Israeli authorities. The use of such data by the OECD is without prejudice to the status of the Golan Heights, East Jerusalem and Israeli settlements in the West Bank under the terms of international law.

© OECD 2025



Attribution 4.0 International (CC BY 4.0)

This work is made available under the Creative Commons Attribution 4.0 International licence. By using this work, you accept to be bound by the terms of this licence (<https://creativecommons.org/licenses/by/4.0/>).

Attribution – you must cite the work.

Translations – you must cite the original work, identify changes to the original and add the following text: In the event of any discrepancy between the original work and the translation, only the text of original work should be considered valid.

Adaptations – you must cite the original work and add the following text: This is an adaptation of an original work by the OECD. The opinions expressed and arguments employed in this adaptation should not be reported as representing the official views of the OECD or of its Member countries.

Third-party material – the licence does not apply to third-party material in the work. If using such material, you are responsible for obtaining permission from the third party and for any claims of infringement.

You must not use the OECD logo, visual identity or cover image without express permission or suggest the OECD endorses your use of the work. Any dispute arising under this licence shall be settled by arbitration in accordance with the Permanent Court of Arbitration (PCA) Arbitration Rules 2012. The seat of arbitration shall be Paris (France). The number of arbitrators shall be one.

Acknowledgements

This paper was written by Jan-Peter Kleinhans, Mario Alejandro Nieves, Sara Romaniega Sancho, and Charles-Édouard van de Put of the Directorate for Science Technology and Innovation (STI). It was edited by Shai Somek, and benefited from the valuable insights, comments, and inputs of Guy Lalanne, Jens Lundsgaard, Audrey Plonk, Lea Samek, and Filipe Silva. The authors gratefully acknowledge feedback provided by the Semiconductor Informal Exchange Network participants as well as Antoine Berthou, Chiara Criscuolo, and Gallia Daor in earlier versions of this document. The authors also thank Anaisa Goncalves and Andreia Furtado for their support, and Joaquim Guilhoto and Antton Haramboure for constructing and providing part of the data used in this paper.

Abstract

The semiconductor value chain is highly complex, with globally distributed production processes and tightly interconnected segments. This paper identifies the key segments, its actors, inputs, and datasets, offering a comprehensive view of the semiconductor ecosystem. It examines each segment to identify bottlenecks and vulnerabilities. The analysis of cross-country linkages within the semiconductor value chain reveals a high concentration of critical inputs in specific regions, the specialisation of economies in certain segments and growing trade dependencies. Policymakers should consider actions to mitigate vulnerabilities and enhance the resilience of the semiconductor value chain, including through efforts to diversify supply chains and through sharing of information on market and policy developments. Strengthened international collaboration and knowledge sharing will be essential to address vulnerabilities and manage potential supply chain disruptions.

Table of contents

Acknowledgements	3
Abstract	4
Executive summary	7
1 Introduction	8
2 Overview of the semiconductor value chain	9
Definitions and key segments	9
Actors and inputs across different segments	11
3 Using trade data and ICIO tables to help map the value chain	19
Data and methodologies	19
Exports, specialisation and concentration by product type	21
Main dependencies in the semiconductor value chain	33
Input-output relationships and links across industries and main economies	34
4 Conclusion	37
Endnotes	38
References	40
Annex A. Available data and data sources	45
Annex B. Efforts to map the semiconductor value chain	49
Annex C. Inter-Country Input-Output (ICIO) tables	53
Annex D. List of HS codes	56
Endnotes	59

FIGURES

Figure 2.1. Key segments of the semiconductor value chain	9
Figure 2.2. Nodes in the semiconductor value chain	11

6 | MAPPING THE SEMICONDUCTOR VALUE CHAIN

Figure 2.3. Annual revenue shares in 2023 from different geographical regions for selected companies	13
Figure 3.1. Overview of the different segments in the semiconductor value chain	20
Figure 3.2. Global exports market share and RCA for raw materials	22
Figure 3.3. Export concentration for raw materials	23
Figure 3.4. Global exports market share and RCA for inputs for wafer fabrication	24
Figure 3.5. Export concentration for inputs for wafer fabrication	24
Figure 3.6. Global exports market share and RCA for inputs for silicon wafers	25
Figure 3.7. Export concentration for silicon wafers	25
Figure 3.8. Global exports market share and RCA for inputs for foundry inputs	26
Figure 3.9. Export concentration for foundry inputs	27
Figure 3.10. Global exports market share and RCA for inputs for manufacturing equipment	28
Figure 3.11. Export concentration for manufacturing equipment	29
Figure 3.12. Global exports market share and RCA for inputs for chips	30
Figure 3.13. Export concentration for chips	31
Figure 3.14. Global exports market share and RCA for photosensitive semiconductor devices	32
Figure 3.15. Export concentration for photosensitive semiconductor devices	32
Figure 3.16. Number of trade dependencies across economies	33
Figure 3.17. Number of trade dependencies across segments	34
Figure 3.18. Gross output in the semiconductor industry, 2012-2018	34
Figure 3.19. Upstream and downstream connected sectors, 2018	35
Figure 3.20. Main upstream and downstream economies in the semiconductor industry, 2018	36
Figure 3.21. Downstream flows of the semiconductor industry, 2018	36
 Figure A C.1. Final goods flow in the electronics sector, 2018	 54

TABLES

Table A D.1. List of HS codes associated to raw materials	56
Table A D.2. List of HS codes associated to inputs for wafer fabrication	56
Table A D.3. List of HS codes associated to silicon wafers	57
Table A D.4. List of HS codes associated to foundry inputs	57
Table A D.5. List of HS codes associated to manufacturing equipment	57
Table A D.6. List of HS codes associated to final products (chips)	58
Table A D.7. List of HS codes associated to photosensitive semiconductor devices	58
 Table A A.1. Detail on data sources	 47

Executive summary

The semiconductor value chain is highly complex, characterised by globally distributed production processes and numerous interlinked segments. Comprehensive understanding and effective risk monitoring and management of this value chain are crucial but require addressing significant data gaps. This paper provides evidence on dependencies and vulnerabilities in the semiconductor value chain to help inform policymakers in considering policies to strengthen its resilience. Working together with stakeholders to diversify supply sources and enhancing international collaboration as well as knowledge sharing can help reduce trade dependencies and manage potential supply chain disruptions.

More specifically, this paper provides a first mapping of the semiconductor value chain, focusing on two essential aspects: i) the structure of the semiconductor value chain and ii) its main vulnerabilities and bottlenecks. It identifies key segments, actors, and inputs, and offers a detailed analysis of how the different components and processes within the value chain interact, as part of the complex interdependencies that characterise the industry.

UN Comtrade data used in this paper show that the number of trade dependencies for semiconductors is particularly high, having more than doubled from 2012-14 to 2020-22. Dependencies for raw materials and manufacturing equipment are also high and have increased considerably in the same period. The supply of borates, silicon wafers and various inputs for wafer fabrication is highly concentrated, with several Asian producers playing key roles. Detailed analysis of bilateral trade flows shows that, for numerous products, many economies increasingly rely on the People's Republic of China (hereafter “China”), with limited alternative suppliers. This reliance creates dependencies not only for semiconductors themselves but also for several inputs along the value chain.

Building on OECD Inter-Country Input-Output data, this paper provides a more comprehensive analysis of the concentration of intermediate inputs flows, differentiating between final and intermediate goods (e.g. packaged and unpackaged semiconductors) and identifies whether the production is intended for domestic use or export. Data show that about half of the intermediate inputs used globally in the semiconductor industry are components sourced from within the industry itself, highlighting a high degree of intra-industry dependency. In parallel, approximately one-third of intermediate outputs are sold to other semiconductor firms for further processing. This dynamic highlights the layered and iterative nature of semiconductor production. In other words, the industry is deeply interconnected: firms rely on one another not just for components, but also for successive processing stages—sending parts back and forth as they are gradually assembled, refined, or integrated into final semiconductor products. Additionally, over half of all semiconductor intermediate outputs go to China, highlighting both the concentration of production and the vulnerabilities in the event of supply chain disruptions.

Enhancing the resilience of the semiconductor value chain by mitigating vulnerabilities is an important topic for policymakers. The OECD supports these efforts by analysing different segments of the value chain, by providing policymakers with data to help identify bottlenecks, monitor market imbalances and manage disruptions in the value chain, and by supporting international dialogue and exchange on best policy practices.

1 Introduction

During its inaugural year in 2023, the OECD Semiconductor Informal Exchange Network (hereafter the Network) discussed the importance of understanding and monitoring the semiconductor value chain, as well as managing value chain risks. The Network highlighted data challenges and affirmed the need for a comprehensive understanding of the semiconductor value chain in detail. To support this objective, the Network developed a taxonomy of semiconductor products and production facilities (OECD, 2024^[1]) as a first step, with the longer-term goal of identifying and measuring key upstream and downstream activities in the semiconductor value chain.¹

Mapping the semiconductor value chain provides critical insights into the interdependencies between production segments and the flow of inputs and outputs throughout the manufacturing process. This understanding is essential for addressing data gaps, identifying vulnerabilities, and informing policy decisions, in line with the objectives of the Network (OECD, 2023^[2]).

This paper explores two key aspects of the semiconductor value chain: The first examines its structure to better understand the different segments, linkages, inputs and outputs. A thorough understanding requires first identifying the inputs (and their sources) and outputs (and their destinations) at different segments of the semiconductor manufacturing process. This understanding also helps identify other industries beyond semiconductors where the resilience of the whole ecosystem, i.e. the network of firms, workforce, processes, and sectors involved in or dependent on semiconductor production, could be strengthened.

The second aspect focuses on identifying bottlenecks and vulnerabilities in the semiconductor value chain. A deeper understanding of these vulnerabilities is crucial for policymakers to manage risks and respond effectively to potential disruptions. The industry may also benefit from better information about vulnerabilities further upstream in the value chain (i.e., beyond the first tier of suppliers). This entails important data challenges. The analysis in this paper builds on the previous identification of key segments and their interdependencies, leveraging available data sources.

While detailed, granular data are available for upstream and front-end manufacturing, data availability is comparatively limited for back-end manufacturing, as well as for Research and Development (R&D) and Design. The analysis builds on highly disaggregated UN Comtrade data to quantify input-output relationships within the value chain, which are further complemented by OECD Inter-Country Input-Output (ICIO) data to derive key information such as intermediate consumption. There are nevertheless important data aggregation challenges that limit the analysis, notably for semiconductor inputs. Further work will investigate each segment in more detail.

The remainder of this paper is structured as follows. Section 2 identifies key nodes, inputs and actors of the semiconductor value chain. Section 3 describes the value chain using trade data and ICIO tables. Section 4 concludes.

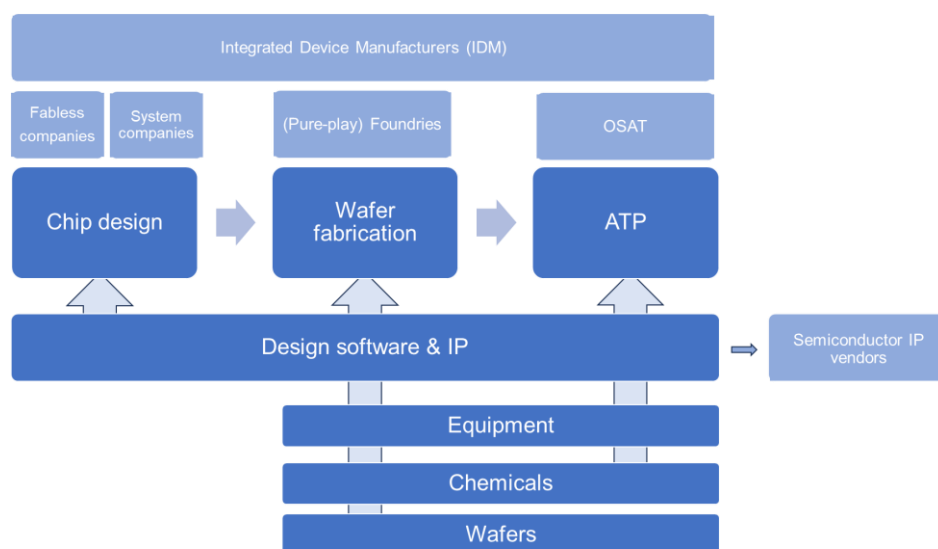
2 Overview of the semiconductor value chain

This section describes the semiconductor value chain in detail. For each key segment, the process is briefly explained, and the inputs (material and equipment) are outlined. A short list of key market players is provided for each.

Definitions and key segments

The semiconductor value chain is extremely complex and relies on a vast network of different types of actors with varying degrees of interaction. As shown in Figure 2.1, the semiconductor value chain can be segmented into three major process steps: design, wafer fabrication and assembly, test and packaging (ATP). Each of these process steps depends on distinct inputs, such as wafers (i.e. a thin slice of a semiconductor material, usually silicon, on which electronic circuits are created), equipment and materials, from often highly specialised suppliers. Both the process steps and their suppliers are heavily reliant on R&D, resulting in the semiconductor industry having among the highest R&D margins across industries (Bloom et al., 2020^[3]).

Figure 2.1. Key segments of the semiconductor value chain



Note: The light blue boxes indicate the different business models.

Each process step entails different structural characteristics, including barriers to entry. Chip design is skill- and research- intensive, while wafer fabrication is highly capital-intensive and relies on economies of scale. In contrast, ATP is less automated, making it more labour-intensive, with typically lower added value and

profit margins, compared to wafer fabrication and chip design. Because of these distinct differences between process steps, semiconductor companies have become more specialised, typically focusing on only one of the steps. This specialisation has contributed to the growing complexity and interdependence of the value chain over the last three decades. Below is a brief overview of the different business models that emerged and their relation to the three process steps.

- **Fabless companies** design and sell chips. They outsource front- and back-end manufacturing to foundries (i.e. manufacturing facilities producing chips for other companies) and Outsourced Semiconductor Assembly and Test (OSAT) companies, respectively. Fabless companies have among the highest R&D margins in the semiconductor industry, often spending 25% and more of their revenue on R&D (Kleinhans, 2021^[4]). Most of the fabless chip design companies are headquartered in the United States, such as Nvidia, Qualcomm and AMD.
- **System companies** design chips for their own products. They differ from fabless companies by selling systems (cars, smartphones, or services) rather than individual chips. Like fabless companies, they rely on foundries and OSATs for manufacturing. Examples of system companies with substantial chip design capabilities are Apple, Tesla, Google and Alibaba.
- **Semiconductor Intellectual Property (IP) vendors** design and sell functional blocks (also referred to as “IP cores”) that are used by chip designers to shorten time to market and lower chip design costs. Arm, Cadence, Synopsys and Imagination Technologies are among the largest IP vendors (Esteve, 2024^[5]).
- **(Pure-Play) Foundries** operate wafer fabrication plants (fabs) to provide contract wafer fabrication services to companies that design chips (fabless, Integrated Device Manufacturers (IDM) and system companies). They closely collaborate with IP vendors and Electronic Design Automation (EDA) suppliers (see below) to offer a “rich” IP value chain to their customers.
- **Integrated Device Manufacturers (IDM)** design, produce and sell chips. Traditionally, IDMs handled all three process steps – design, front- and back-end manufacturing – internally. Historically, this was the prevalent business model in the semiconductor industry. However, over the last two decades, IDMs have increasingly outsourced parts of their production, following the so-called “fab-lite” business model. Today, the “fab-lite” IDM model is prevalent, where IDMs decide on a product-by-product basis whether front- and/or back-end manufacturing is done in-house or outsourced. Some IDMs, such as Intel and Samsung, also offer foundry services for contract manufacturing. Unlike Pure-Play Foundries, **IDM-Foundries** use some or all of their fabs for contract manufacturing for external customers and internal production of its own chips.
- **Outsourced Semiconductor Assembly and Test (OSAT)** companies offer contract back-end manufacturing services to external customers.

It is important to understand that the level of vertical integration (across process steps) does not just vary at the company-level but already at the product level, illustrated by the following three examples — see (OECD, 2024^[1]) for an overview of product types and manufacturing process technologies.

Intel, like many other IDMs, has gradually shifted toward a fab-lite business model, outsourcing some elements of both front- and back-end manufacturing to foundries and OSATs (Mujtaba, 2024^[6]). At the same time, the company is expanding its foundry business to offer contract manufacturing to external customers for both front-end and back-end manufacturing. Thus, Intel could be considered a fab-lite IDM, an IDM-Foundry and an OSAT.

Infineon, Texas Instruments, Renesas and STMicroelectronics are examples of fab-lite IDMs in the analog, power, mixed-signal and microcontroller markets. While they may carry out front- and back-end manufacturing of a power semiconductor in-house, these same process steps could be entirely outsourced for microcontrollers. In the case of silicon-carbide (SiC) power semiconductors, some of these companies

are fully vertically integrated, from production of SiC ingots and epitaxy of the SiC wafer to packaging of the final chip (Yole Group, 2022^[7]).

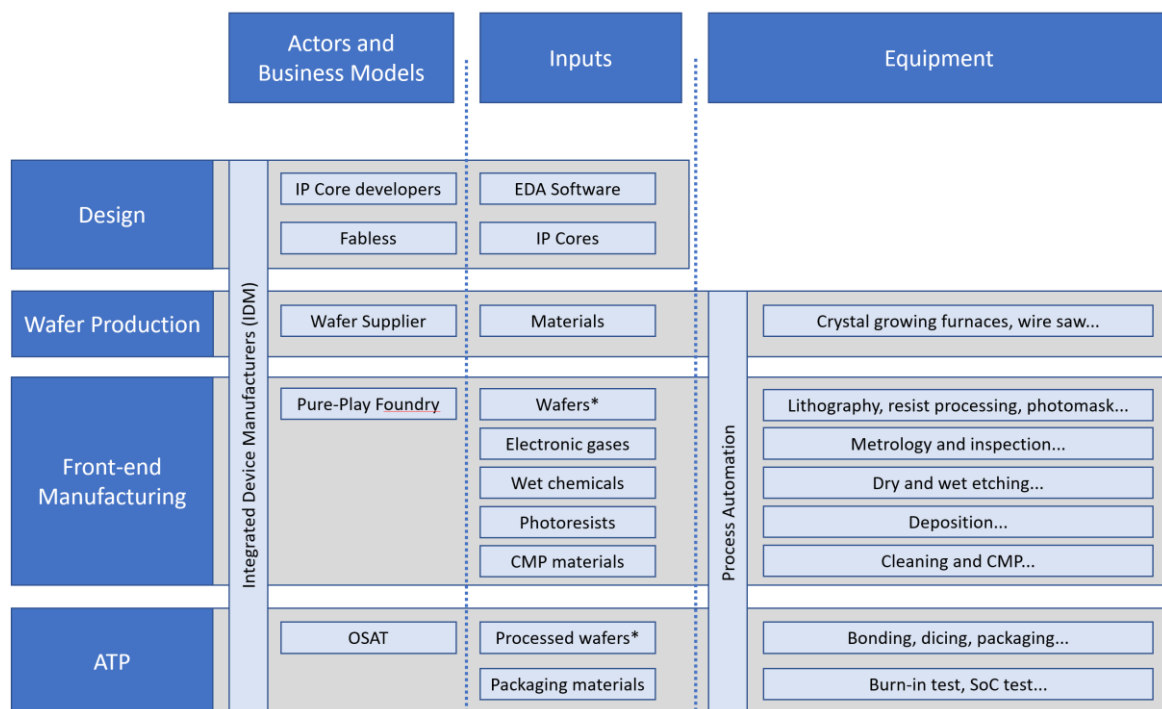
Alibaba, as a system company with their in-house chip design unit T-Head, is developing chips for the company's cloud business (Loghin, 2024^[8]). Additionally, Alibaba is developing IP cores and selling chips to customers (Chen et al., 2020^[9]). Thus, Alibaba's business could be characterised as a system company, a fabless company and an IP vendor.

These examples illustrate that semiconductor companies often do not have one fixed role in the value chain but engage in various activities along the value chain. The following subsection will provide a more detailed explanation of the different segments (process steps and their inputs) of the semiconductor value chain.

Actors and inputs across different segments

This section describes the semiconductor manufacturing process from design to ATP. Figure 2.2 shows how the key nodes of the value chain can be categorised into different groups of actors. An overview of the activities of each actor is provided along with an inventory of inputs and key market players therein.

Figure 2.2. Nodes in the semiconductor value chain



Note: * input from previous segment.

Source: OECD, based on CSET (2021^[10]) and ETO Supply Chain Explorer (2022^[11]).

Chip Design

Semiconductor development begins with digital chip design and an increasing number of companies are involved in chip design today. Regardless of whether fabless companies, system companies or IDMs designing a semiconductor, all rely on three key inputs: Electronic design automation (EDA) tools, IP cores and process design kits (PDK) (Ferguson, 2018^[12]).

- **IP cores** are pre-designed, reusable functional blocks used in chip designs. They are crucial to reducing development time, ensuring reliability, and meeting the stringent performance and power requirements of modern chips. Developing a high-quality IP block, such as a Central Processing Unit (CPU) core or a SerDes interface, requires years of expertise and domain-specific knowledge. Furthermore, (upfront) licensing fees and royalty structures (often paid on a per-unit basis) create ongoing financial and contractual dependencies between chip designer and IP vendor.
- **EDA tools** are specialised software used by engineers to bring together semiconductor designs using IP cores and custom designs. It allows them to design, simulate and verify the design. Modern chip design processes involve more than 40 individual steps and most chip designers “mix and match” EDA tools to select the best-in-class tool per step (Nenni, 2022^[13]). The costs of designing a chip vary widely depending on the feature size of the process node, the complexity of the design as well as license costs for EDA tools and IP cores (Shilov, 2023^[14]).
- **PDKs** ensure that semiconductor designs align with the capabilities of the foundry where the chip will be taped out. IDMs and foundries develop PDKs for their process nodes that contain all the design rules of a node, such as minimum distances between functional elements. While the IDM uses the PDK only internally, foundries provide the PDKs to their customers. The EDA software then uses the PDK’s design rules to ensure that the design created can be manufactured on this process node. Foundries regularly improve their PDKs and there is often close collaboration between EDA tool vendors, such as Cadence and Synopsys, and foundries.

Innovation in semiconductor design starts through novel designs of chips and IP cores. R&D is particularly crucial for optimising and overcoming the design challenges of creating the most efficient semiconductors within the current physical limitations of the fabrication process, while considering the planned end use (logic, memory, etc.).

R&D plays a critical role in all steps of semiconductors production. Innovations in electronics, material science, and miniaturisation, among others, can significantly impact the entire value chain. This can, for example, manifest in improved designs, manufacturing processes or equipment for both state-of-the-art and next-generation semiconductor technologies. The fabrication process itself is an important area of ongoing R&D in companies and across the value chain (e.g., through institutions such as Interuniversity Microelectronics Centre in Leuven, Belgium). Therefore, many key technology partners and input providers throughout the semiconductor value chain invest heavily in R&D.

Landscape

The current landscape of companies designing chips is quite diverse. Traditionally IDMs, such as NXP, Renesas and Texas Instruments, as well as fabless companies, such as AMD, Broadcom and MediaTek, were designing chips. While this remains the case, system companies, such as Alibaba, Apple and Google, increasingly invest in their internal chip design (Ma, 2022^[15]). As a result, it is challenging to map the global chip design landscape and compare chip design capacities of different economies solely based on data from IDMs and fabless companies.

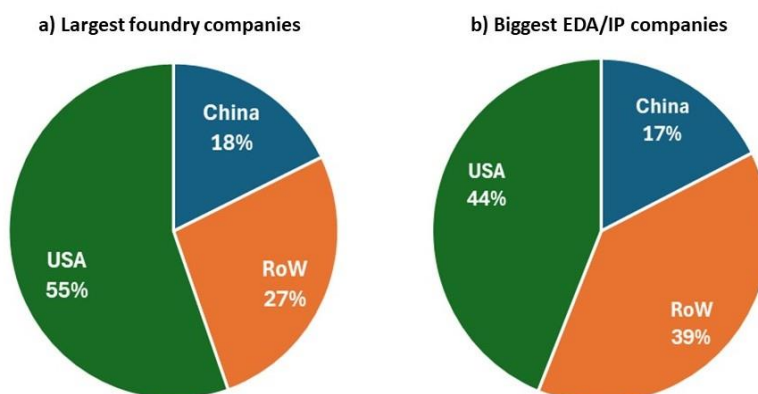
One way to better understand chip design capacity of different economies is to examine revenue shares by geography from the largest foundries. This approach captures all design activities from fabless and system companies, as well as some activities from fab-lite IDMs, which also outsource parts of their manufacturing.

Figure 2.3, left panel, shows the revenue shares by geographical region from TSMC, Globalfoundries, UMC, SMIC and Hua Hong in 2023. Together, these five foundries account for 80% of the global foundry market. Customers headquartered in the United States accounted for roughly 55% of sales, while customers in the People’s Republic of China (hereafter “China”) generated the second-highest sales, at 18%. The high share of foundry customers headquartered in the United States is not surprising, given that

many of the largest fabless and system companies, such as AMD, Apple, Nvidia and Qualcomm, are headquartered there (BCG, SIA, 2022^[16]).

Figure 2.3. Annual revenue shares in 2023 from different geographical regions for selected companies

Panel a: % of foundry revenue by geography; panel b: % of EDA/IP revenue by geography



Note: The left panel includes the following foundry companies: TSMC, UMC, SMIC, GlobalFoundries, and Hua Hong Semiconductors which make up approximately 80% of the market share. The value for China and the United States may be underreported due to geographical region categorisation by each of the companies. A further breakdown of “Rest of the World” is not possible given that regional categories widely differ between these five foundries. The right panel includes the following EDA/IP companies: Cadence, Synopsys Inc., and Arm.

Source: OECD based on TSMC (2024^[17]), UMC (2023^[18]), SMIC (2024^[19]), GlobalFoundries Inc. (2023^[20]), and Hua Hong Semiconductor Limited (2023^[21]) (left panel) and Cadence Design Systems (2023^[22]), Synopsys Inc (2023^[23]), and Arm Holdings plc (2023^[24]) (right panel)

Another perspective is to examine the geographical revenue distribution of EDA and IP vendors. Since both markets are highly concentrated, looking at geographical revenue shares of the largest players provides insight into which economies spend the most on EDA and IP, indicating the most commercially active chip design value chain. Figure 2.3, right panel, shows the revenue share by geographical region for Cadence, Synopsys and Arm. Together, these companies account for more than 60% of the global EDA market and 70% of the IP market (Zeoli, 2024^[25]; Esteve, 2024^[5]). This figure closely resembles the left panel, with the notable difference that the share of the United States is lower, at 44% instead of 55%. One explanation is that other economies, such as Japan, Korea and the European Union, are largely dominated by IDMs, which also license EDA software and IP from these vendors.

Tools and equipment

- **EDA Tools:** the market is concentrated in the United States through three companies, namely Cadence, Siemens EDA (formerly Mentor Graphics), and Synopsys (Kleinhans, 2022^[26]).
- **Core IP:** the market is split between the United States and the United Kingdom through companies such as Arm, Synopsys, Cadence, SST and Imagination Technologies (Clarke, 2024^[27]).

Wafer production

Wafers can consist of various materials, such as silicon, silicon carbide (SiC), gallium nitride and diamond. However, silicon is by far the dominant wafer material and the backbone of today's semiconductor manufacturing, especially for logic and memory chips. In 2023, more than 250 million (8-inch equivalent) silicon wafers were shipped (SEMI, 2024^[28]), compared to an estimated 1.7 million SiC wafers (irrespective of diameter) (Huang, 2023^[29]). The production process for different types (materials) of wafers varies, but

the goal remains the same: to achieve high purity, uniformity, and in the long-term an ever-increasing wafer diameter (mks, 2024^[30]).

Fabs typically obtain raw wafers from specialised providers. In order to deliver a raw wafer, the specialised production facility has to produce a cylinder (referred to as a boule or ingot) of pure silicon (or another material). Wafer thin discs are then cut from the cylinder with the help of special machine tools. The level of semiconductivity can then be altered across different parts of the wafer by using ion implanters to embed “dopants”. The latter step can also be performed in the fab as part of chips fabrication. Key inputs to this step include raw semiconducting materials, typically silicon, which are usually refined in a separate facility, as well as furnaces and machine tools for forming and slicing the ingot, implanters and dopant substances for treating the wafers.

Landscape

More than 50% of the global silicon wafer market are supplied by two Japanese companies, Shin-Etsu and SUMCO, with other significant players including GlobalWafers from Chinese Taipei and Siltronic from Germany (Siltronic, 2024^[31]). However, the landscape for other types of wafers, such as silicon-carbide, gallium nitride, sapphire or diamond wafers is fundamentally different. While silicon wafers will continue to dominate, especially for logic and memory chip production, alternative wafer materials provide superior physical characteristics for power and radio frequency chips. The production process and corresponding supply chain for these wafers substantially differs from that of silicon wafers.

Material inputs

Polycrystalline silicon, or polysilicon, is ultra refined metallurgical grade silicon created from quartz. The quartz undergoes multiple purification steps to remove as many impurities as possible, resulting in silicon rods. These rods are key inputs for wafer fabrication, as they allow for the creation of an ingot which will eventually be machined into wafers. The process is very similar for germanium wafers.

Other semiconductor materials can be used as substrates, including, but not limited to, silicon carbide, gallium arsenide, aluminium gallium phosphide, and mercury cadmium telluride. Each requires the eponymous elements in high purity. Nevertheless, the vast majority of semiconductor production currently uses silicon wafers. Although the production of most of these raw materials is largely concentrated in China, many of these elements are present across the globe.

Tools and equipment

- **Crystal growing furnaces:** the market is concentrated in a German company called PVA TePla.
- **Crystal machining tools:** the market is concentrated in Japanese companies Accretech, Okamoto, and Disco.
- **Ion implanters:** the market is concentrated in the United States with one Japanese competitor. Main companies include Applied Materials (United States), Nissin Ion (United States), SMIT (Japan), and Axcelis (United States).

Source: (Center for Security and Emerging Technology, 2021^[10])

Fabrication (front-end manufacturing)

Semiconductor fabrication facilities (or fabs) take raw wafers and create integrated circuits (chips) on them through a multistep process. This typically involves a cycle of polishing, deposition, resist application, lithography, etching, ion implantation, and resist removal, repeated according to the number of layers required for the final chip (Halbleiter, 2024^[32]). This description is vastly simplified and does not reflect the

complexity and sensitivity of the process. Indeed, the most advanced fabrication facilities are some of the most complex and expensive production facilities, taking years to build. See the OECD semiconductor taxonomy (OECD, 2024^[1]) for further information on semiconductor types and production facilities.

While semiconductor fabrication itself is important, it relies on a wide array of inputs both in equipment and materials. Many of these inputs are specific to semiconductor fabrication and are highly specialised. A prime example are Extreme Ultraviolet (EUV) lithography machines produced by the Dutch company ASML. Modern semiconductor production depends on various lithography tools, such as wet and dry deep ultraviolet lithography tools. For the most advanced production processes at 5nm and below, several layers of a chip can only be produced with EUV equipment. Nonetheless, “older” deep ultraviolet (DUV) lithography equipment is still required for the other layers (Kleinhans and Lee, 2024^[33]). Furthermore, for photomasks production and advanced packaging, other types of lithography equipment, such as X-ray lithography and electron-beam lithography, are essential.

This step of the semiconductor manufacturing process requires the most inputs in terms of equipment and materials, with wafers requiring different specialised tools and inputs as they are processed. Throughout its journey in the fab, a wafer is constantly moved and monitored, requiring specific process controlling and wafer handling equipment.

Landscape

It is important to understand that wafer fabrication is highly diversified, and market analyses need to consider differences in node density, process technology and business models, at the very least. For instance, a modern power semiconductor cannot be built on a process node for high-performance processors or on one for memory chips. Furthermore, some chip types are mainly built in an IDM or a foundry business model. Commodity memory, such as DRAM and NAND, is solely built by IDMs, including Samsung, SK Hynix, Micron, Kioxia and YMTC. These IDMs are also among the companies with the highest wafer capacity, as competition in commodity memory is driven by economies of scale. In contrast, most of today’s smartphone System on Chip (SoC) and Artificial Intelligence (AI) accelerators are designed by fabless and system companies and then manufactured by foundries.

Material inputs

Photomasks: The material and functioning of the photomask depend on the type (or generation) of lithography tool. They allow transferring the pattern onto chips using photolithography. While DUV lithography tools use transmissive masks, often made of glass substrates which are coated with opaque materials (typically chromium), EUV lithography uses reflective masks consisting of multiple layers. Photomask production is either done by the fab internally or by merchant mask shops. It starts with a mask blank that is then processed to create the pattern of each layer of the chip. According to the Center for Security and Emerging Technology (2021^[10]), the production of transmissive masks (DUV lithography) is not particularly concentrated, and the raw material components are only moderately concentrated, even if chromium exports are concentrated, South Africa being the largest exporter (US Geological Survey, 2024^[34]).

Photoresists: these are chemicals that are deposited on the wafers that undergo a chemical reaction when exposed to the light passing through the photomask. This reaction lays the groundwork for etching, which is performed on areas where the photoresist has dissolved. Each generation of lithography uses photoresists that are specifically tuned to the wavelength of the light source. The shorter the wavelength, the more advanced and specialised the resist chemistry must be to meet the increased demands of finer feature sizes. Photoresists are organic materials typically made up of resin, sensitiser, and solvent. Japan is the dominant player in the photoresist market (Center for Security and Emerging Technology, 2021^[10]).

Wet processing chemicals: these chemicals are used for different purposes and process steps. For example, wet processing chemicals for etching are used after lithography to transform solid material into liquid to etch the chip layout. Wet processing chemicals are also used during the various cleaning steps to remove residues. Examples of wet processing chemicals are sulfuric acid, isopropyl alcohol, ammonium hydroxide, phosphoric acid, nitric acid, hydrochloric acid, and acetic acid. Some of these require ultra-high levels of purity and therefore are only produced by a limited number of plants.

Specialty gases: gases are used throughout the semiconductor fabrication process as dopants, dry etchants, in the dry etching process or for protection from atmospheric exposure. These include common gases like nitrogen, hydrogen, and carbon dioxide, electronics specialty gases, such as nitrogen trifluoride, tungsten hexafluoride, hydrogen chloride, ammonia, disilane, germane, high purity carbon dioxide, and nitrous oxide; or noble gases including neon, argon, helium and krypton. The level of concentration of production of these gases varies. Some are widely available while others are produced by only a few specialised companies (SIA, 2021^[35]).

Chemical mechanical planarisation (CMP) slurries: a uniform surface for lithography requires wafers to be planarised using a polishing pad and a slurry, which contains abrasives and chemicals. These include rare earths such as cerium. The production of these slurries and pads is concentrated in Japan (Center for Security and Emerging Technology, 2021^[10]).

Sputtering targets: these are used for Physical Vapor Deposition (PVD) to apply a thin film through the sputtering method which consists of ejecting material from the target on to the wafer. The most commonly used sputtering targets in semiconductor manufacturing are made of aluminium, titanium, tantalum and copper. The choice of target material depends on the type of chip and desired properties. Japan and the United States are the main producers of targets for the industry (Stanford Advanced Materials, 2023^[36]).

Tools and equipment

A large variety of chipmaking equipment is needed in modern wafer fabrication. In addition to equipment for the major process steps – lithography, etching, deposition –, equipment is needed for cleaning, CMP, process automation and metrology and inspection. To understand the landscape, competitive positions and dependencies, these broad categories need to be further broken down. For example, wafer fabrication depends on several different types of deposition equipment in different process steps but also on the type of chip that is built. While Applied Materials is leading the market for Physical Vapor Deposition (PVD) equipment (Castellano, 2023^[37]), ASM and Tokyo Electron are the market leaders in Atomic Layer Deposition (ALD) (Fund, 2024^[38]). A similar situation exists in other chipmaking equipment categories, such as etching, bonding and cleaning equipment, where technology leadership depends on the specific subcategory. Thus, the following list of equipment suppliers should only be seen as a broad overview.

- **Advanced photolithography equipment:** the market is primarily dominated by companies from the Netherlands and Japan, including ASML (Netherlands), Canon (Japan), and Nikon (Japan). ASML is the only firm that provides EUV equipment. Notably, equipment suppliers also depend on various highly specialised component and sub-system suppliers. For instance, ASML depends on over five thousand suppliers and builds only around 15% of a lithography machine itself (Kleinhans and Lee, 2024^[33]).
- **Advanced photomasks:** mask blanks are mainly supplied by Japanese companies, such as Shin-Etsu, Hoya and AGC. The production of photomasks by merchant mask shops is mainly led by companies from Japan and the United States, such as Dai Nippon and Photronics respectively.
- **Deposition tools:** the market is shared between the United States (Applied Materials, Lam Research), Japan (Tokyo Electron), and the Netherlands (ASM).
- **Etching tools:** dry etching (using gases or plasma in a vacuum to remove material) and wet etching (using liquid chemicals) are the two general types of etching equipment used in wafer

fabrication. While wet etching tools are mainly used for cleaning and residual film removal, dry etching is indispensable to build the structures of a chip. There are several different dry etching technologies and different companies are leading in the respective technologies (Huff, 2021^[39]). Lam Research (United States), Applied Materials (United States) and Tokyo Electron (Japan) are among the leading companies in the dry etch tool market (Tsutsumi, 2024^[40]).

- **Wafer cleaning tools:** Screen (Japan), Tokyo Electron (Japan) and Lam Research (United States) accounted for 75% of the wafer cleaning equipment market in 2022 (Hars, 2022^[41]). Since 2020, Chinese companies, such as ACM Research and Naura also gained market shares (Wen, McKay and Ma, 2022^[42]).
- **Resist processing tools:** used to coat, develop and bake the photoresist on the wafer before and after the lithography process step. Leading suppliers are Japanese companies Tokyo Electron and Screen.
- **Wafer bonders and aligners:** joining two or more wafers either permanently, to form stacked structures, or temporarily, where a support wafer is attached to aid in processes like thinning or alignment before being de-bonded. Leading companies are EV Group (Austria), SÜSS MicroTec (Germany), Tokyo Electron (Japan) and Applied Materials (United States).
- **Automated material handling systems:**
The market for photomask handlers is split between the United States and Japan through Brooks and Daifuku, respectively. Wafer handlers are mainly supplied by the Japanese companies Daifuku and Rorze.
- **Process monitoring:** main companies include Keysight Tech, Keithley Instruments in the United States.

Source: (Center for Security and Emerging Technology, 2021^[10])

Assembly, test and packaging (back-end manufacturing)

In this step, the processed wafers are cut into individual dies, i.e. the bare (silicon) substrate that contains the integrated circuit. These dies are then packaged individually into protective frames surrounded by a resin shell. The frame allows the final chips to be soldered onto printed circuit boards (PCB). Throughout this process the chips are rigorously tested to ensure that there are no defects and that they are functioning properly. The chips can be mounted onto a PCB which is shipped to a different assembly facility while in some cases the circuit board is assembled into the final product in the same facility. Collectively these process steps are called ATP or back-end manufacturing.

There is an ever-increasing variety of packaging technologies, and chip designers typically choose a specific packaging technology based on performance requirements, cost constraints, form factor and reliability. The most common categories of packaging technologies are wire bonding, flip chip, wafer-level packaging, system-in-package (SiP) and 3D packaging (Burkacky, Kim and Yeom, 2023^[43]). Packaging technologies are steadily evolving and IDMs, foundries and OSATs offer increasingly specialised packaging approaches targeted for specific applications, such as high-performance computing, automotive and mobile chips (Vardaman, 2021^[44]). For example, advanced packaging techniques such as 2.5D and 3D packaging are increasingly popular, particularly for mobile applications and high-performance computing. These techniques involve stacking multiple dies or integrating different functions onto a single package, improving performance and reducing the space needed (VerWey, 2022^[45]).

Advanced packaging also makes mapping the ATP landscape increasingly complex since Pure-Play foundries, IDM-foundries and (traditional) OSATs now offer advanced packaging processes. The reason is that advanced packaging depends on more sophisticated equipment, better cleanrooms and in general a more controlled environment, blurring the line between front- and back-end manufacturing (Patel, 2022^[46]).

The conventional ATP process requires many distinct tools and equipment to process the finished wafer. In terms of inputs, the assembly and packaging steps usually require more mechanical equipment and material inputs while the testing steps require specialised electronic equipment.

Landscape

In ATP, the segment is split between OSATs offering contract back-end manufacturing and IDMs that do ATP internally. However, as mentioned before, since most IDMs today follow a fab-lite business model, they decide on a product-by-product basis whether to outsource packaging to an OSAT or foundry.

Global revenues from OSAT companies reached USD 44 billion in 2022, with the largest OSAT companies (by revenue) being ASE Group (Chinese Taipei), Amkor (United States), JCET (China) and Tongfu Microelectronics (China) (IDC, 2023^[47]). When focusing specifically on advanced packaging, Intel, TSMC and Samsung are also among the leading players (Yole Group, 2023^[48]).

Material inputs

The most critical inputs are finished wafers from fabs (IDMs or foundries). Traditional packaging materials tend to have less stringent specifications, making them more widely available due to the lower technological challenges of production. Given the concentration of ATP activities in Southeast Asia, the production of these packaging components is also somewhat concentrated in the region, particularly in Japan.

Lead frames: are metal structures that support and secure chips while carrying signals from the chip to the wiring board. They are made from thin metal plates that have been etched or stamped to desired patterns. They are usually made from copper, but alternative materials are copper alloys and iron-nickel alloys, which can be gold or silver plated.

Packaging substrates: chips must be protected and are therefore encased in substrates. The choice of substrates depends on the type of usage for the chip and the desired properties. The main types of packaging substrate materials are plastic, ceramic, metal, organic, and composites (Cadence Design Systems, n.d.^[49]). Advanced packaging substrates are on the cutting edge of development, including glass substrates for heterogeneous integration.

Bonding wires: connect the chip to the packaging and, in some cases, to other electronics. The main materials for these wires are copper, aluminium, gold, and silver.

Die-attach materials: are used to attach the chip (die) to a substrate or package (lead frame, ceramic substrates, PCBs, etc.). Their core functions are mechanical adhesion, electrical insulation, and thermal management. The three main categories of materials used for die-attaching are epoxy resin, metal alloys and ceramics (Ayodele, 2023^[50]).

Tools and equipment

- **Dicing tools:** concentrated in Japan through Disco and Accretech.
- **Integrated assembly tools:** these integrate multiple packaging steps and systems. Among the leading companies are ASM Pacific Technologies (ASMPT) (Singapore) and Besi (Netherlands).
- **Testing equipment:** split between Japan (Advantest), the United States (Teradyne) and Korea (UniTest).

Source: (Center for Security and Emerging Technology, 2021^[10])

3

Using trade data and ICIO tables to help map the value chain

The analyses in this section show that export of chips have doubled since 2012, with most segments having experienced an increase, especially in 2021 and 2022. The market for chips is dominated by several Asian producers that are also leaders in other segments of the value chain, namely in silicon wafers, inputs for wafers, manufacturing equipment and foundry inputs. Concentration and bilateral trade dependencies have been on the rise since 2012. At the global level, the supply of borates is highly concentrated, with the Republic of Türkiye (hereafter "Türkiye") as a key supplier. Similarly, the supply of silicon wafers and various inputs for wafer fabrication is highly concentrated with Japan and other Asian producers such as Korea and Chinese Taipei playing key roles. At the bilateral level, many economies seem to rely heavily on a main importer, with limited alternative suppliers, thereby creating dependencies not only for chips themselves but for several inputs along the value chain.

This section includes a brief overview of the data and methodologies used, followed by an overview of exports, specialisation and concentration by product type; an assessment of the main dependencies in the semiconductor value chain; and an analysis of input-output relationships and cross-sectoral and cross-economy linkages.

Data and methodologies

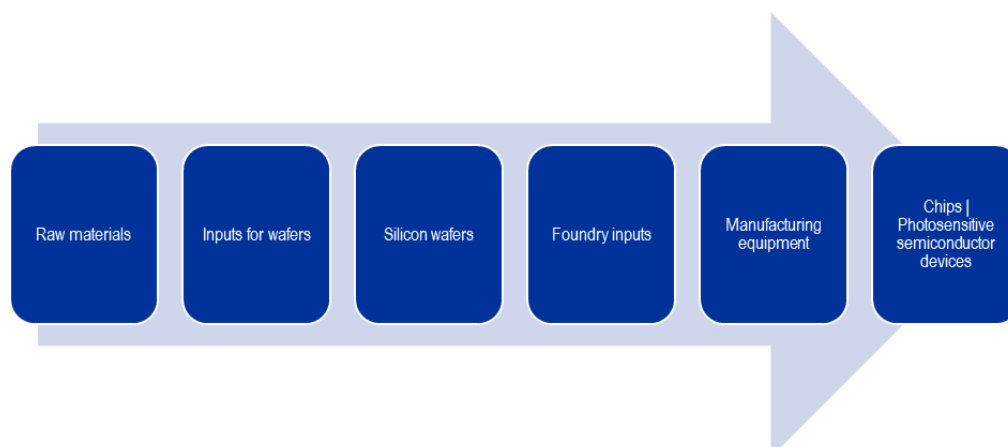
The first part of this analysis focuses on the trade of inputs necessary in each segment of the value chain, using primarily data from the United Nations (UN) Comtrade Database (Comtrade). Comtrade allows monitoring global trade by trading partner and product at the 6-digit level. While more granular information (at the 8- or 10-digit level) is available for certain economies,² the analysis is limited to the 6-digit level for comparability.³

While production data at the product level is available for certain economies, a global database similar to Comtrade does not currently exist for production data. This means that it is challenging to obtain a consistent view of production for domestic use at the product level, limiting the analysis of value chain dynamics on a global scale.

Previous research by Haramboure et al. (2023^[51]),⁴ Bonnet and Ciani (2023^[52]), OECD (2019^[53]) and desk research helped identify key products for each segment of the semiconductor value chain (Figure 3.1). Differentiating by segment, the subsequent step involves analysing the cross-country evolution of global exports and examining the market concentration and trade dependencies.

This section also provides a semiconductor-augmented application of the ICIO tables.⁵ Unlike trade data, ICIO tables provide a sectoral understanding of each economy's contribution to Global Value Chains (GVCs). This allows for a deeper understanding of how different economies contribute to semiconductor production and intermediate material flows, offering richer insights than trade data alone.

Figure 3.1. Overview of the different segments in the semiconductor value chain



The analysis of exports allows assessing trends and understanding their underlying drivers at both the economy and product levels. Additionally, understanding an economy's specialisation at different segments of the supply chain is crucial to determine trade dependencies and detect possible supply disruptions. One useful tool for this assessment is the Revealed Comparative Advantage (RCA). The RCA indicator compares an economy's export share in a particular semiconductor segment to the global share, thereby providing evidence into the economy's specialisation. The level of concentration in each market is quantified using the Herfindahl-Hirschman Index (HHI). In addition, further criteria are used to identify import dependency of a given OECD Member country to a partner economy for a particular product, building on the methodology developed by Kowalski and Legendre (2023^[54]). Further details on the methodology can be found in Box 1.

Box 1. Methodological Framework

Revealed Comparative Advantage (RCA)

The RCA compares an economy's export share in a particular segment to the global export share and is expressed as follows:

$$RCA_{Ai} = \frac{\frac{X_{Ai}}{\sum_{j \in P} X_{Aj}}}{\frac{X_{wi}}{\sum_{j \in P} X_{wj}}} \quad \text{Equation 1}$$

where X_{Ai} represents the exports of economy A in segment i , $\sum_{j \in P} X_{Aj}$ represents the total exports of goods for economy A,⁶ X_{wi} is the world's exports of segment i , and $\sum_{j \in P} X_{wj}$ accounts for the world's total exports of all goods. Therefore, a RCA value above 1 indicates a comparative advantage.

Herfindahl-Hirschman Index (HHI)

The HHI measures market concentration by summing the squared market shares of each economy for a given good. A three-year average of trade flows is used to mitigate the impact of noise and outliers. The index equals one in the case of a monopoly and converges to 0 with a large number of atomistic suppliers. Thus, the closer the index is to zero, the lower the concentration.

The interpretation of the HHI level varies by context. Vicard and Wibaux (2023^[55]) and Bonnet and Ciani (2023^[52]) deem a product concentrated when the index exceeds 0.4. Following Kowalski and Legendre

(2023^[54]), this paper defines markets with a HHI between 0.15 and 0.25 as moderately concentrated, and those with a HHI greater than 0.25 as highly concentrated.

Trade dependencies

To assess bilateral trade dependencies, this paper follows the approach developed by Kowalski and Legendre (2023^[54]). Three conditions need to be met to consider an OECD Member country *i* dependent for its supply of good *k*. First, gross imports of good *k* by economy *i* need to be overall highly concentrated (HHI >0.4). Second, the imports from a partner economy account for a high share (>10%) of *i*'s total imports of good *k*. Third, other OECD Member countries' import share for good *k* in economy *i* is less than 20%. To avoid labelling an economy with a high domestic production as dependent, two additional conditions apply. The ratio of exports to imports of good *k* in economy *i* must be below the 90th percentile of the distribution and imports from economy *i* of good *k* need to amount to USD 1 million or more.

However, these analyses provide only a partial view of the ecosystem value chain and its main risks for several reasons. First, concentration of trade flows does not take into account other aspects that are key to determine supply chain risks, such as the criticality of the input for the production process and the availability of substitutes. Second, trade data provide an incomplete view of the supply chain. For instance, some goods are counted twice (e.g. a chip fabricated in country X, packaged in country Y and used in country Z will appear twice in the same category), whereas production for domestic use is not taken into account. Third, the granularity of the data is not sufficient to accurately capture the specifics of the semiconductor industry. For instance, although the HS classification covers over 5 thousand goods, some categories are too broad, bundling several types of non-substitutable chips, or a wide range of machines, not all of which are relevant for the semiconductor value chain.

Exports, specialisation and concentration by product type

This section analyses exports and indicators of specialisation and concentration for key product types, namely Raw materials, Inputs for wafer fabrication, Silicon wafers, Foundry inputs, Manufacturing equipment, Chips, and Photosensitive semiconductor devices.

Raw materials

Table A D.1 provides a comprehensive list of HS codes associated with the raw materials used in the analysis. This list includes various raw materials critical in the semiconductor production process, such as natural borates, rare gases, silicon and specialised compounds like silicon carbide.

Exports of raw materials decreased from 2017 to 2020 (from USD 13.8 to USD 10.9 billion) but rebounded in 2021 and 2022, surpassing USD 18 billion. In 2022, the United States was the largest exporter, followed by Germany and China. Notably, China's market share in global exports increased from 7.8% in 2012 to 14.5% in 2022 (Figure 3.2, left panel). Silicon (HS code 280461) accounted for approximately 33% of total raw material exports, with Germany and the United States being its principal exporters. However, these results should be interpreted carefully given that trade data only measures inter-country flows and exclude domestic production. According to the United States Geological Survey (2024^[34]), China was the largest producer of silicon materials in 2023, responsible for 69% of global estimated production.

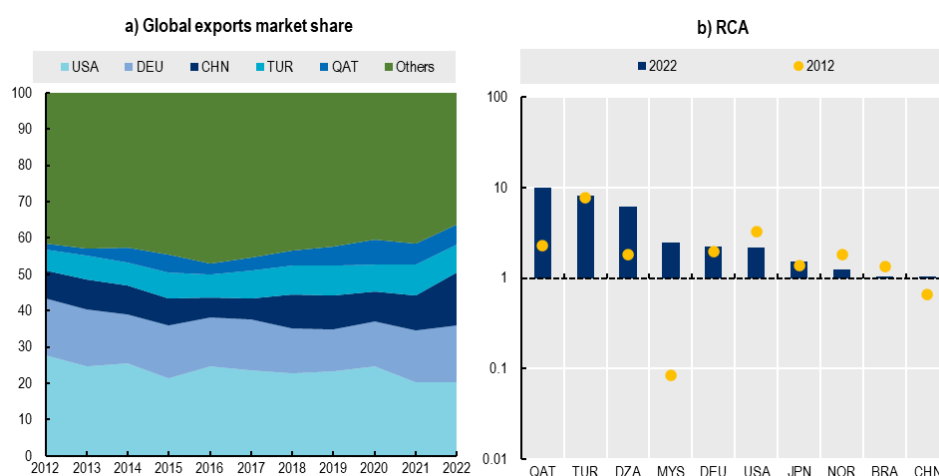
Gallium and germanium are crucial elements in semiconductor manufacturing, particularly for wafer fabrication. In 2022, China represented over 30% of the global market share for raw metals classified under HS code 811292, which includes gallium and germanium, as well as other metals like hafnium, rhenium, and vanadium. However, when examining gallium and germanium individually, China's significant role

becomes more evident. According to the United States Geological Survey (2024^[34]), China produced over 98% of the world's gallium in 2023 and continued to be the leading global producer and exporter of germanium. While comprehensive global germanium refinery production data for 2023 is limited, the most recent figures from 2020 indicate that China accounted for 68% of global germanium production. Additionally, despite a lack of export data for August and September 2023 following China's export control measures introduced in August, exports in the year until September still rose by 34% compared to the same period in 2022 (US Geological Survey, 2024^[34]).

Qatar and Türkiye are the economies with the highest RCA (Figure 3.2, right panel). In 2022, Türkiye was the largest exporter of natural borates (HS code 252800) and borates, such as disodium tetraborate (refined borax), excluding anhydrous (HS code 284019). Meanwhile, Qatar was the largest exporter for noble gases, excluding argon (HS code 280429).

Figure 3.2. Global exports market share and RCA for raw materials

Panel a: % of global exports; panel b: RCA indicator - logarithmic scale

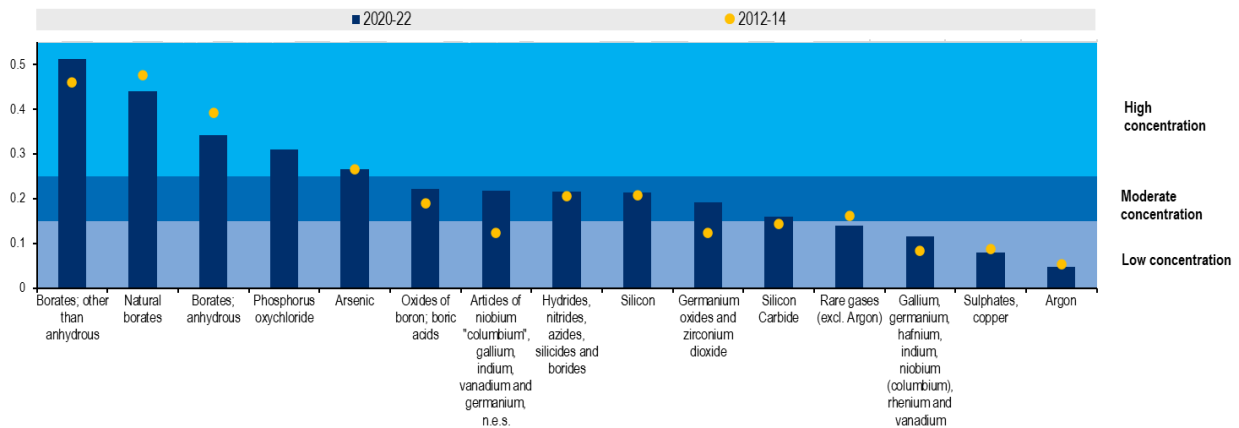


Note: The left panel shows the five largest exporters in 2022. The right panel shows the ten economies with the highest RCA scores for which exports account for at least 1% of total exports in 2022. The latest observations available are for 2022. Data for phosphorus oxychloride prior to 2017 is not available.

Source: OECD calculations based on UN Comtrade database.

As shown in Figure 3.3, the export concentration is higher for the three categories of borates, in particular disodium tetraborate (refined borax), excluding anhydrous (HS code 284019), which has the highest concentration in 2020-22. Among all the products analysed in this section, it exhibits the highest degree of export concentration. In the most recent period, the share of Turkish exports in the global market surpassed 60%.

Figure 3.3. Export concentration for raw materials
Herfindahl-Hirschman Index



Note: Data for phosphorus oxychloride prior to 2017 is not available.
Source: OECD calculations based on UN Comtrade database.

Inputs for wafer fabrication

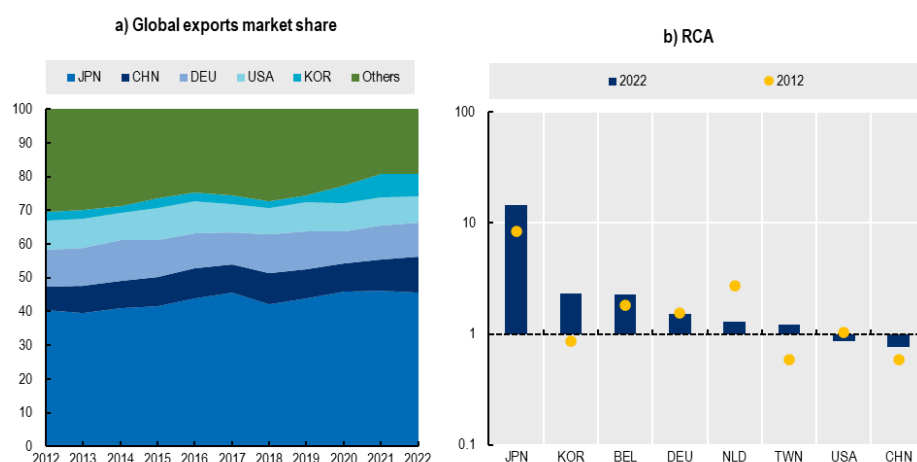
Table A D.2 provides the HS codes associated with the inputs needed for wafer fabrication, including various photographic products, e.g., plates, films and related goods, used in the semiconductor industry.

The exports of inputs for wafers followed a W-shaped pattern between 2012 and 2022 with a peak in 2018 and two troughs in 2016 and 2020. Similar to raw materials, exports rebounded notably in 2021 and 2022, reaching approximately USD 12 billion by 2022. Japan is the largest exporter, accounting for over 40% of global exports in 2022 and significantly surpassing the next four major exporters, namely Germany, China, Korea, and the United States (Figure 3.4, left panel). Notably, 65% of exports in this category were linked to exports of photographic goods (HS code 370790), where Japan is the world leader.

Japan's specialisation in this segment is also marked by its high RCA score, which exceeded 10 in 2022. Among the ten economies with the highest score, only six exhibit an RCA above 1 in 2022 (Japan, Korea, Belgium, Germany, the Netherlands, and Chinese Taipei). Of these, only the Netherlands has seen a decline in its comparative advantage since 2012 (Figure 3.4, right panel).

Figure 3.4. Global exports market share and RCA for inputs for wafer fabrication

Panel a: % of global exports; panel b: RCA indicator - logarithmic scale



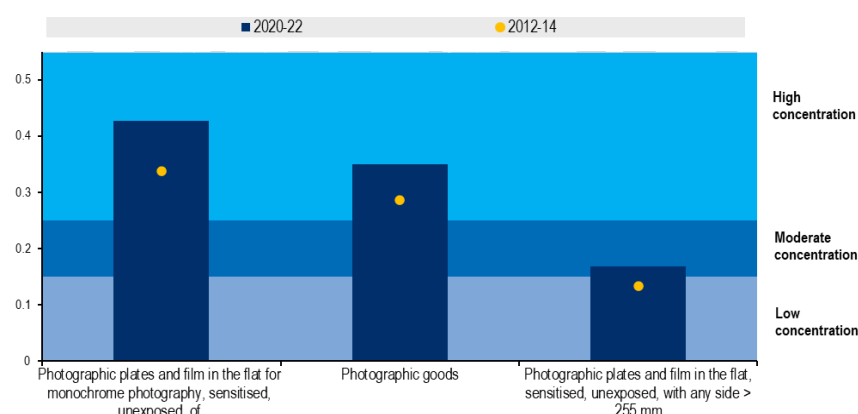
Note: The left panel shows the five largest exporters in 2022. The right panel shows the ten economies with the highest RCA score for which exports account for at least 1% of total exports in 2022. The latest observations available are for 2022.

Source: OECD calculations based on UN Comtrade database.

Among the products analysed under inputs for wafers, two exceed the threshold for high concentration, with photographic plates (HS code 370199) being the most concentrated product. For this product, Japan accounted for 64% of exports in the most recent period. Concentration also increased for the other two products. While photographic plates with any side > 255 mm (HS code 370130) stayed below the threshold for moderate concentration in 2012-14, this product is now considered moderately concentrated (Figure 3.5).

Figure 3.5. Export concentration for inputs for wafer fabrication

Herfindahl-Hirschman Index



Source: OECD calculations based on UN Comtrade database.

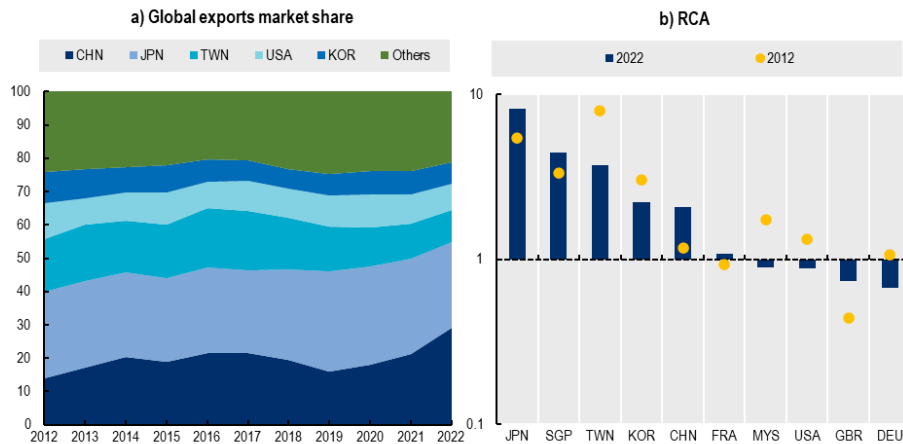
Silicon wafers

Silicon wafers are identified in the HS classification using the code listed in Table A D.3. Exports of silicon wafers stagnated between 2018 and 2020 but rebounded in 2021 and 2022, growing by 46% from around USD 13 billion in 2012 to USD 24 billion in 2022. China was the leading exporter in 2022, accounting for

29% of global exports, followed by Japan, Chinese Taipei, the United States and Korea (Figure 3.6, left panel). In terms of RCA, Japan leads, followed by Singapore, Chinese Taipei, and Korea. Among these economies, Korea and especially Chinese Taipei experienced a significant decline in relative specialisation. Conversely, Japan, Singapore and China saw an increase in their RCA indicator, with Japan showing the largest increase (Figure 3.6, right panel).

Figure 3.6. Global exports market share and RCA for inputs for silicon wafers

Panel a: % of global exports; panel b: RCA indicator - logarithmic scale



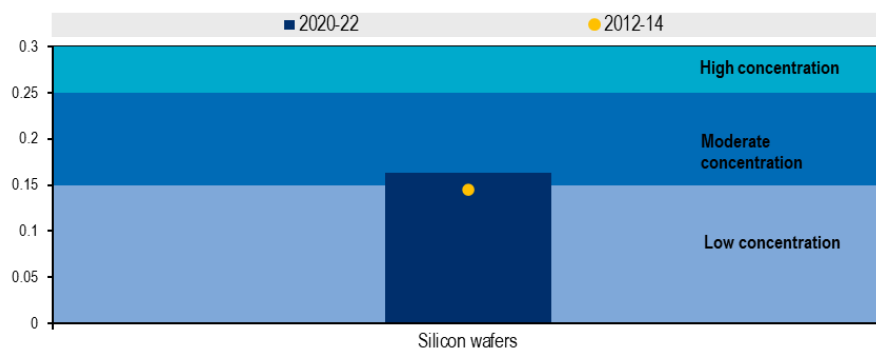
Note: The left panel shows the five largest exporters in 2022. The right panel shows the ten economies with the highest RCA score for which exports account for at least 1% of total exports in 2022. The latest observations available are for 2022.

Source: OECD calculations based on UN Comtrade database.

As shown in Figure 3.7, exports of silicon wafers remained slightly above the threshold for moderate concentration during the 2020-22 period, with only marginal changes compared to the 2012-14 period. More recently, Japan accounted for 29% of the global exports, followed by China with 19% and Chinese Taipei with 12%.

Figure 3.7. Export concentration for silicon wafers

Herfindahl-Hirschman Index



Source: OECD calculations based on UN Comtrade database.

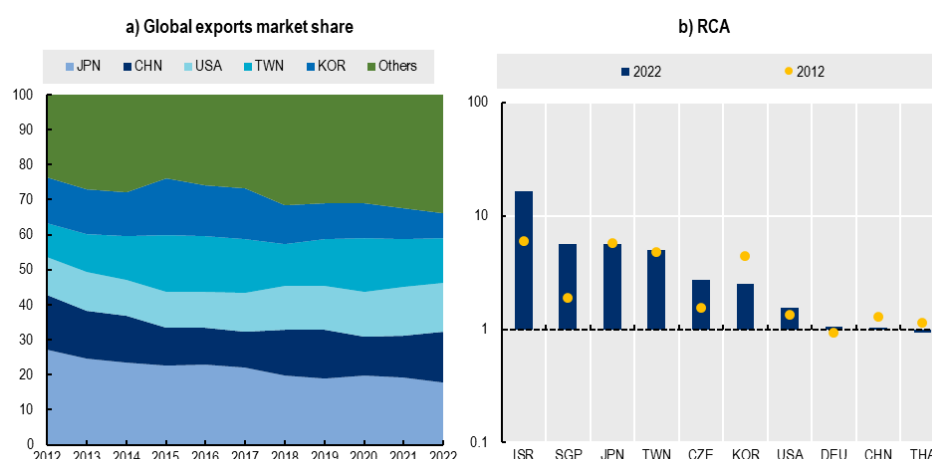
Foundry inputs

Table A D.4 provides the list of HS codes related to foundry inputs used in the analysis, including optical components, such as polarising material sheets, unmounted and mounted lenses, prisms, mirrors, and optical filters. It also lists codes for specialised equipment like electron and proton microscopes, along with their accessories, and identifies optical instruments used for inspecting semiconductor devices.

The export trajectory of foundry inputs steadily increased from 2016, with stagnation between 2018 and 2020, before rebounding significantly in 2021 and 2022 to surpass USD 42 billion. In 2022, Japan was the main exporter, followed by Chinese Taipei and the United States. Between 2012 and 2022, the market shifted, with the United States and Chinese Taipei each gaining about 3 percentage points in global export market share, reaching 14% and 13%, respectively. Meanwhile, Korea and Japan saw declines of around 6 and 10 percentage points, bringing their shares to 7% and 18%, respectively (Figure 3.8, left panel).

Figure 3.8. Global exports market share and RCA for inputs for foundry inputs

Panel a: % of global exports; panel b: RCA indicator - logarithmic scale



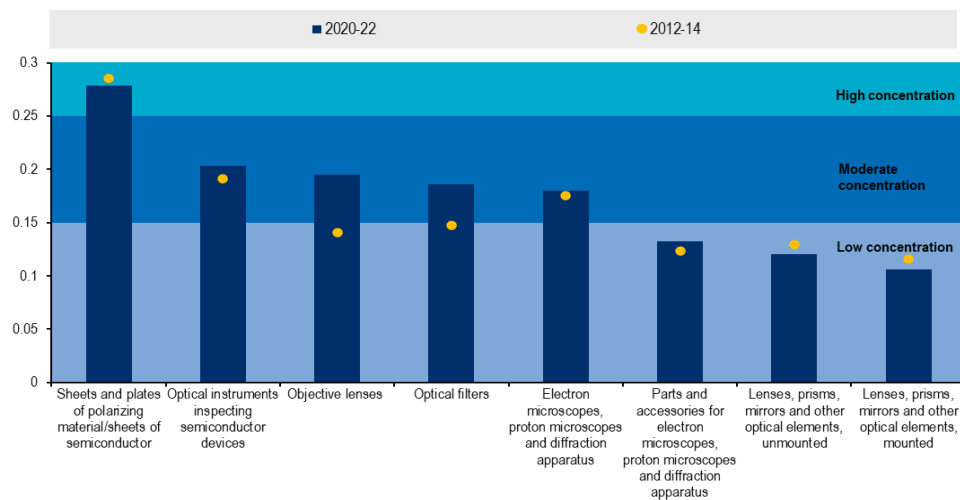
Note: The left panel shows the five largest exporters in 2022. The right panel shows the ten economies with the highest RCA score for which exports account for at least 1% of total exports in 2022. The latest observations available are for 2022.

Source: OECD calculations based on UN Comtrade database.

In 2022, the most traded product was unmounted lenses, prisms, mirrors, and other optical elements (HS code 900190), followed by optical instruments for inspecting semiconductor devices (HS code 903141) and sheets and plates of polarising material/sheets of semiconductor (HS code 900120). China led exports of unmounted lenses, while the United States was the top exporter of optical instruments. Japan emerges as the primary exporter of sheets and plates of polarising material/sheets, contributing approximately 40% of the total exports in 2022. Nonetheless, in terms of RCA, Israel exhibited the highest specialisation. Israel's RCA indicator saw the highest increase among the ten economies with the highest specialisation in this segment (Figure 3.8, right panel). In 2022, exports of optical instruments inspecting semiconductor devices (HS code 903141) from Israel amounted to 20.5% of global exports.

As illustrated in Figure 3.9, exports in the foundry inputs segment remained highly concentrated for sheets and plates of polarising material/sheets of semiconductor (HS code 900120). For this product, the main three exporters (Japan, Korea and Chinese Taipei) accounted for 84% of the total exports in 2020-22. In contrast, for the less concentrated product (mounted lenses, prisms, mirrors and other optical elements, HS code 900290) the top three exporters only accounted for 45% of the global exports.

Figure 3.9. Export concentration for foundry inputs
Herfindahl-Hirschman Index



Source: OECD calculations based on UN Comtrade database.

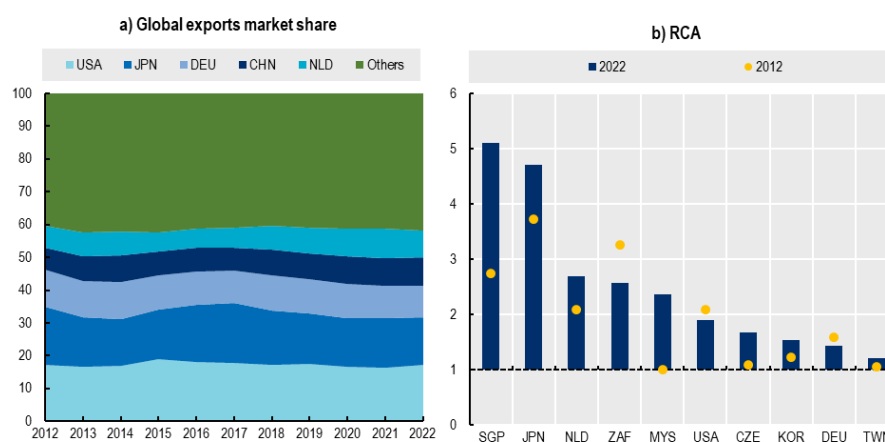
Manufacturing equipment

Table A D.5 includes a comprehensive list of HS codes for various types of manufacturing equipment used in semiconductor production. It includes codes for machinery such as air or vacuum pumps, heat-exchange units, and filters for liquids and gases. The table also details specific equipment for semiconductor manufacturing, such as machines for making semiconductor boules or wafers. Additionally, it includes equipment for measuring or checking semiconductor wafers, electrical quantities, and controlling instruments.

Exports of manufacturing equipment have been intricately connected with the surge in the demand for chips. From 2012 to 2022, exports of manufacturing equipment increased by more than 100%, reaching over USD 270 billion by 2022. By economies, Japan and the United States take the lead in the market, with Germany, China and the Netherlands also playing significant roles. In 2022, the most traded product, accounting for 28% of total exports, was machines and apparatus of a kind used solely or principally for the manufacture of semiconductor devices or of electronic integrated circuits (HS code 848620). Japan accounted for 27% of this products' exports, followed by the Netherlands with 22%. In recent years, Singapore and other emerging players, such as Malaysia, have increased their specialisation in this market. Singapore has already surpassed established players like Japan and the Netherlands in terms of their RCA score. However, while Japan and the Netherlands have seen an increase in their RCA score since 2012, the RCA indicators for the United States and Germany have decreased (Figure 3.10, right panel).

Figure 3.10. Global exports market share and RCA for inputs for manufacturing equipment

Panel a: % of global exports; panel b: RCA indicator - logarithmic scale

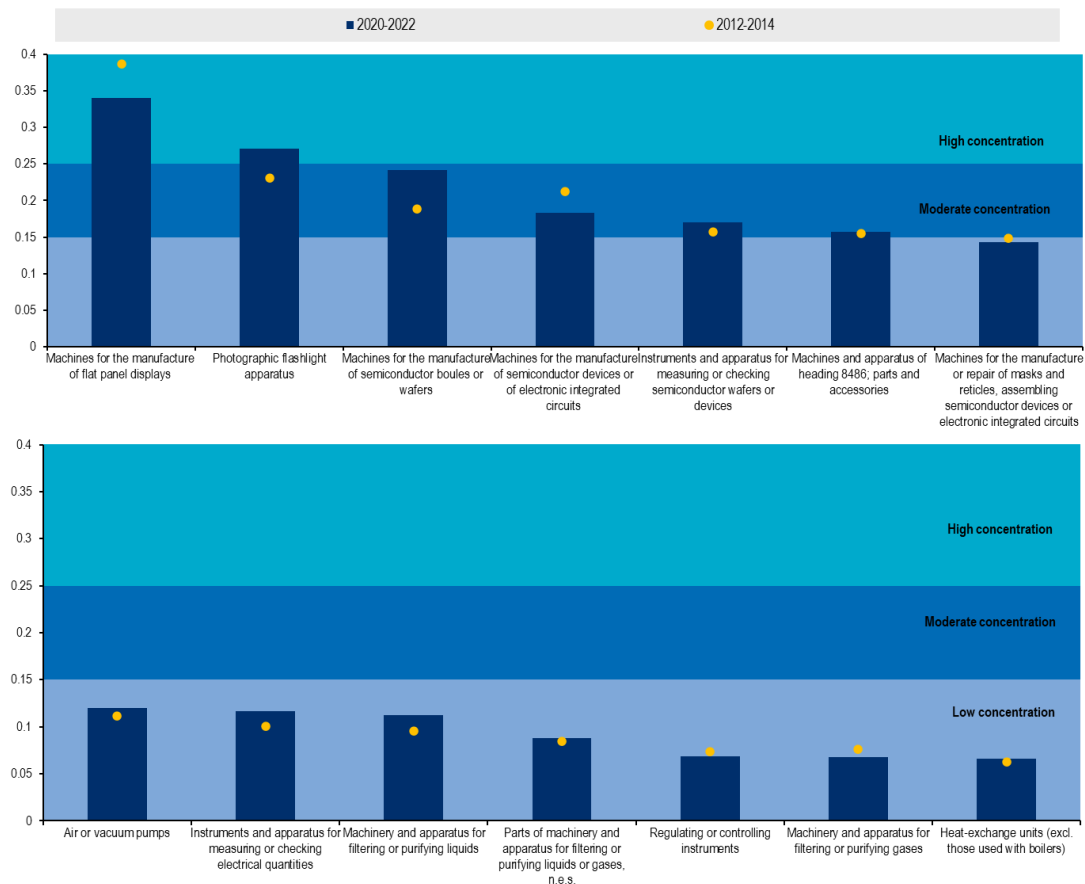


Note: The left panel shows the five largest exporters in 2022. The right panel shows the ten economies with the highest RCA score for which exports account for at least 1% of total exports in 2022. The latest observations available are for 2022.

Source: OECD calculations based on UN Comtrade database.

Exports in the manufacturing equipment segment remained highly concentrated for machines and apparatus used in the manufacture of flat panel displays (HS code 848630) and photographic flashlight apparatus (HS code 900699). Concentration has decreased for the former, while it has increased for the latter (Figure 3.11). In the latest period, the top three exporters (Japan, Korea and Chinese Taipei) of flat panel displays manufacturing equipment accounted for 95% of total exports.

Figure 3.11. Export concentration for manufacturing equipment
Herfindahl-Hirschman Index



Source: OECD calculations based on UN Comtrade database.

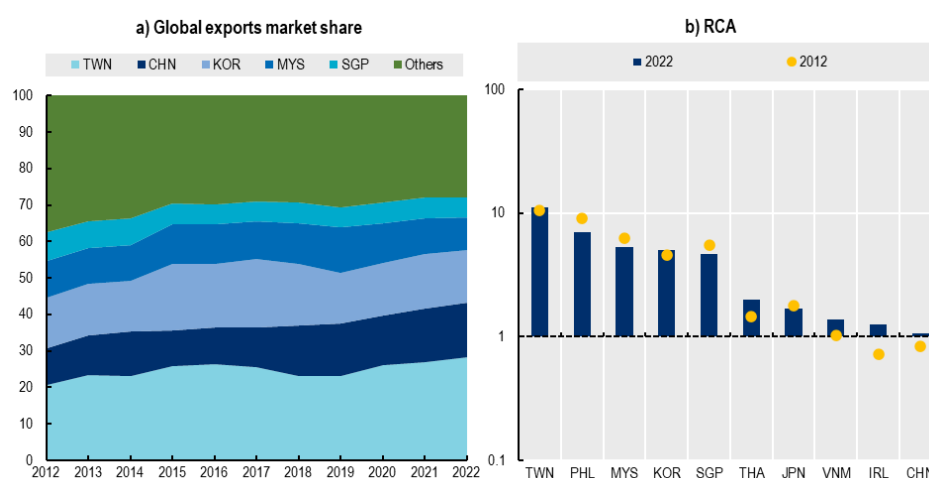
Chips

Table A D.6 provides a list with various types of chips, such as diodes, transistors (both low and high dissipation rates), thyristors, and piezo-electric crystals. The list includes electronic integrated circuits, specifying processors, controllers, memories, amplifiers, and other circuit components. It also covers parts for these integrated circuits and various semiconductor media like flash memory cards and smart cards. Additionally, it includes parts for electrical capacitors and resistors.

Exports of chips have been on the rise since 2012.⁷ In 2012, exports of chips accounted for around USD 600 billion, and by 2022, had increased to more than USD 1 300 billion. The market is dominated by Asian producers, namely China, Korea, Malaysia, Singapore and Chinese Taipei. These five economies accounted for 72% of total exports in 2022, with Chinese Taipei representing 28.1% of global exports (Figure 3.12, left panel). In terms of relative specialisation, Chinese Taipei, the Philippines, and Malaysia exhibit the highest comparative advantage. Chinese Taipei and Ireland saw the highest gain in RCA during this period, while the Philippines experienced the largest decline.

Figure 3.12. Global exports market share and RCA for inputs for chips

Panel a: % of global exports; panel b: RCA indicator - logarithmic scale



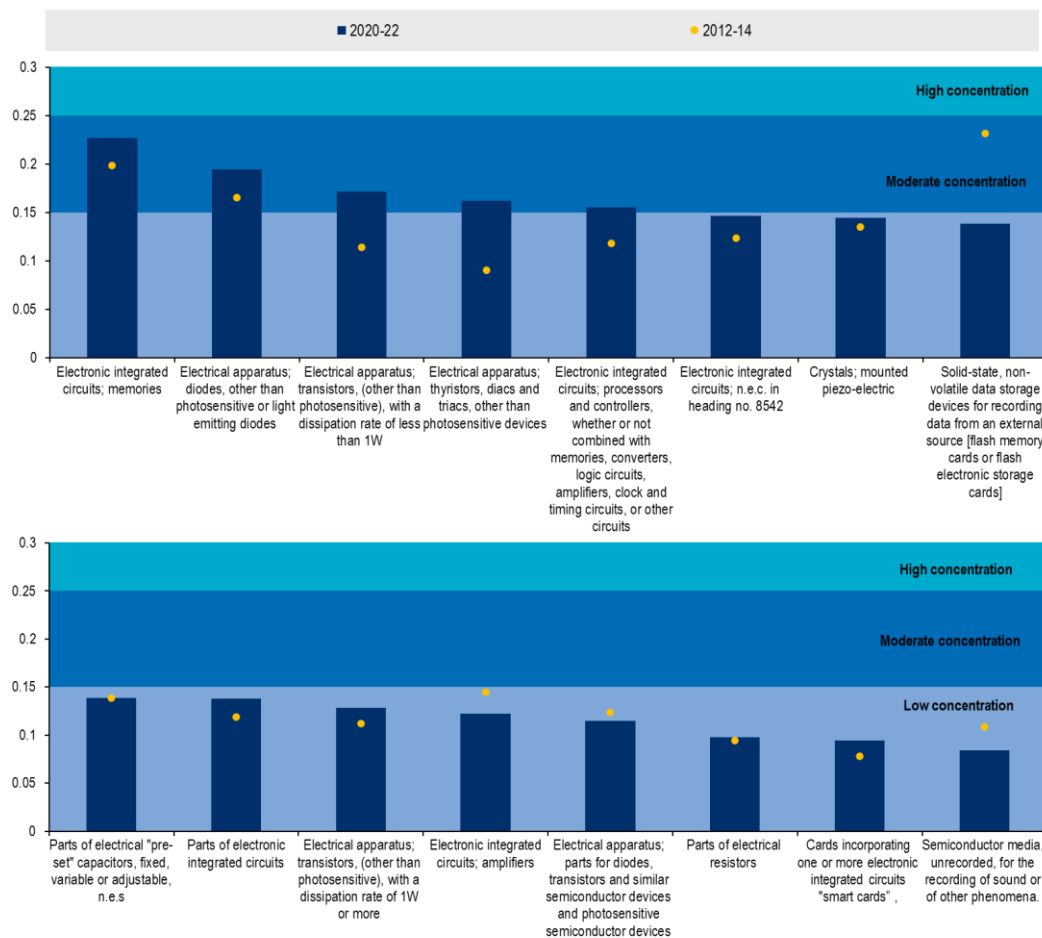
Note: The left panel shows the five largest exporters in 2022. The right panel shows the ten economies with the highest RCA score for which exports account for at least 1% of total exports in 2022. The latest observations available are for 2022.

Source: OECD calculations based on UN Comtrade database.

Exports of all the products categorised as chips remained below the threshold for high concentration (Figure 3.13). Nonetheless, concentration has increased for 12 out of the 16 products, notably for thyristors, diacs and triacs (HS code 854130) and transistors, with a dissipation rate of less than 1W (HS code 854121).

Conversely, concentration decreased for flash memory cards (HS code 852351), where economies such as Korea, Malaysia or the Philippines have gained market share. The share of the top two largest exporters in the period 2012 - 2014 (China and Chinese Taipei) has decreased from 64% in the first period to 42% in the latest period, whereas Korea's share in global exports has increased from 7% to 18% in the same period.

Figure 3.13. Export concentration for chips
Herfindahl-Hirschman Index



Source: OECD calculations based on UN Comtrade database.

Photosensitive semiconductor devices

Despite being made of semiconductor materials, photosensitive semiconductor devices, such as photovoltaic cells and light emitting diodes, are not classified strictly as semiconductors because their primary functions differ significantly from other semiconductor components like transistors and integrated circuits. Photovoltaic cells are primarily used in the solar energy industry, whereas the chips outlined in the previous section are primarily used in electronics and computing to process and control electrical signals. This implies that the production steps and product requirements are notably different. The products categorised as photosensitive semiconductor devices are listed in Table A D.7.

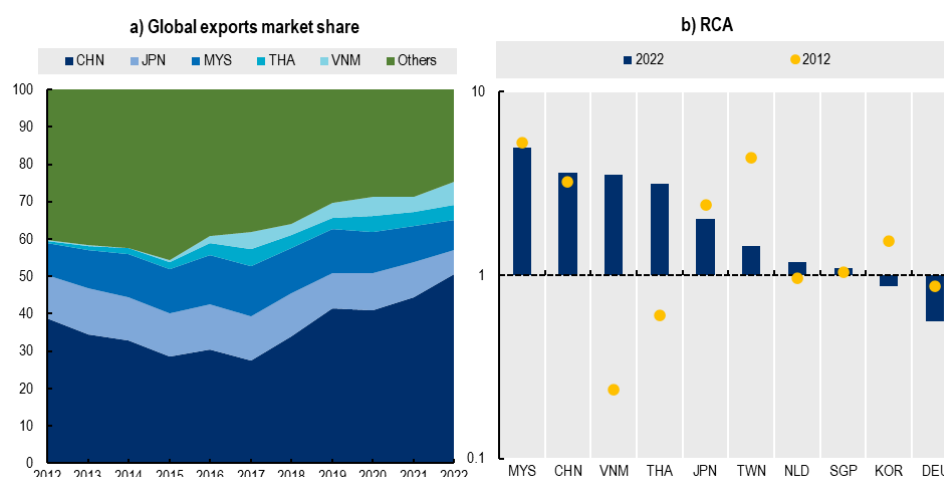
Exports of photosensitive semiconductor devices grew significantly from approximately USD 64 billion in 2012 to over USD 100 billion in 2022, primarily driven by exports of photovoltaic cells; light emitting diodes (HS code 854140). China's influence in this market has grown substantially, with its market share increasing from 38.9% in 2012 to 50.6% in 2022. Emerging economies like Thailand and Viet Nam have also expanded their presence, with Thailand's market share rising from 0.8% in 2012 to 3.9% in 2022, and Viet Nam's increasing from 0.2% to 6.2% over the same period (Figure 3.14, left panel).

The increasing role of these emerging economies is further highlighted by their increasing specialisation. Viet Nam and Thailand saw the highest increases in the RCA indicator, surpassing other Asian producers

such as Chinese Taipei, Singapore, and Korea, as well as European producers like the Netherlands and Germany. However, their RCA scores remain below that of China and Malaysia (Figure 3.14, right panel).

Figure 3.14. Global exports market share and RCA for photosensitive semiconductor devices

Panel a: % of global exports; panel b: RCA indicator - logarithmic scale



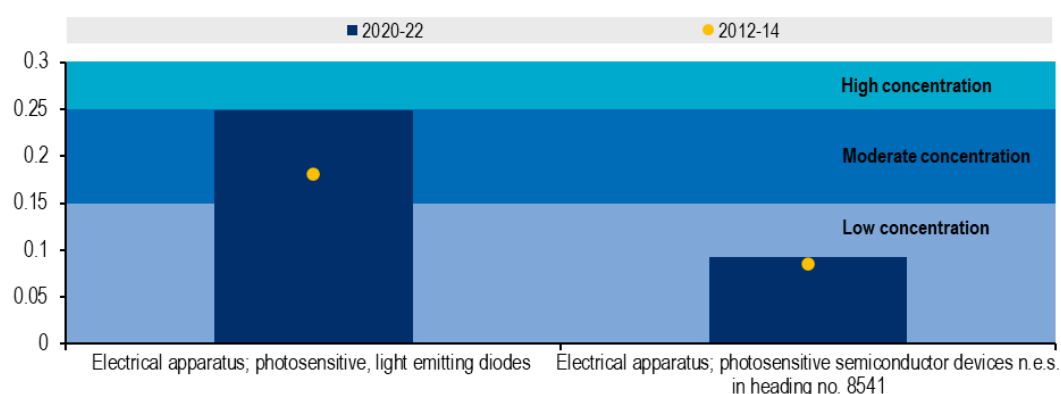
Note: The left panel shows the five largest exporters in 2022. The right panel shows the ten economies with a highest RCA score for which exports account for at least 1% of total exports in 2022. The latest observations available are for 2021 for the HS code 854150 and 2022 for the HS code 854140. The results for 2022 are thus based on the HS code 854140.

Source: Own elaboration based on UN Comtrade database.

Exports of photosensitive semiconductor devices (854140) remained just below the threshold for high concentration in the latest period, showing a significant increase compared to the concentration observed in the initial period (Figure 3.15). In contrast, the concentration of other photosensitive semiconductor devices (854150) has remained consistently low.

Figure 3.15. Export concentration for photosensitive semiconductor devices

Herfindahl-Hirschman Index



Note: The latest observations available are for 2021 for the HS code 854150 and 2022 for the HS code 854140. The results for 2022 are thus based on the HS code 854140.

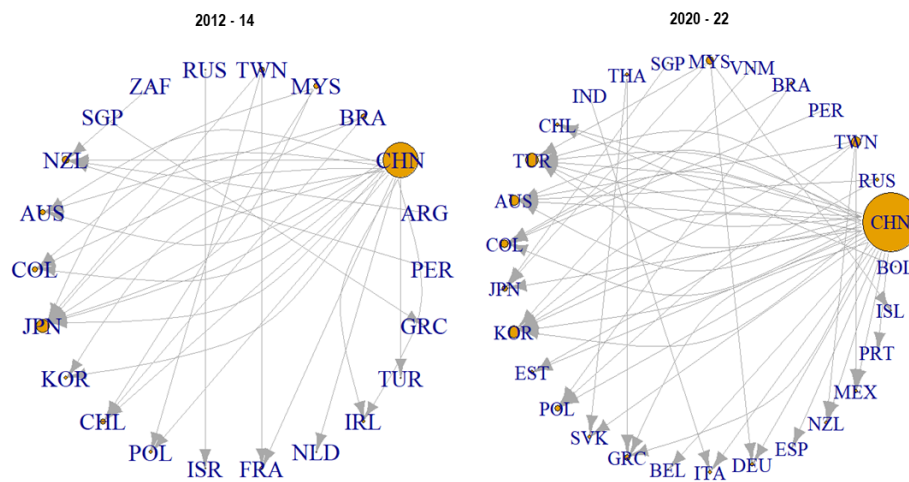
Source: OECD calculations based on UN Comtrade database.

Main dependencies in the semiconductor value chain

The existence of few economies supplying certain goods can induce trade dependencies that may ultimately result in supply bottlenecks, shortages and higher prices. To identify dependencies, the HHI can also be computed for each importer economy and each commodity, revealing whether an economy's imports of a specific product are concentrated in a limited number of economies. Further to the analysis of concentration in the market, additional criteria are used to determine if the economy is trade dependent on a specific product, building on the methodology developed by Kowalski and Legendre (2023^[54]) to determine bilateral trade dependencies (Box 1).

Following this approach, in the latest period (2020-22), 49 dependencies are identified for OECD Member countries, of which 29 correspond to trade flows with China (Figure 3.16). Compared to the first period (2012-14), trade dependencies have increased by 53%, with China's share in total dependencies increased by 3 percentage points, from 56% to 59% (Figure 3.16).

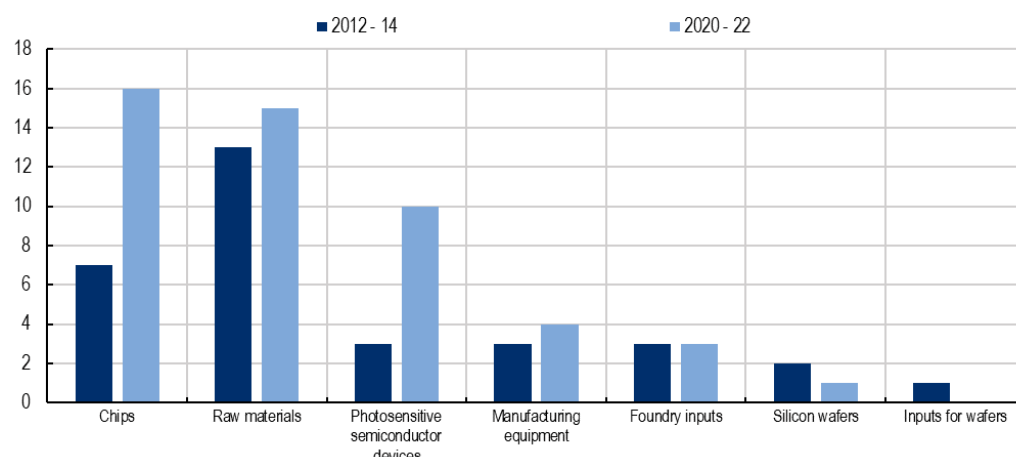
Figure 3.16. Number of trade dependencies across economies



Note: Each arrow represents one dependency at the economy-product level. Node sizes represent degree centrality, indicating the number of direct connections each node has. Note: Dependencies are calculated based on the methodology explained in Box 1.

Source: OECD calculation based on UN Comtrade database.

By segments, dependencies notably increased in the chips segment, rising from seven in the initial period to 16 in the latest period (Figure 3.17). This growth occurred across four out of 16 products in this category, with parts of electrical "pre-set" capacitors (HS code 853290) showing the highest increase in dependencies from one to four. Bilateral trade dependencies for photosensitive semiconductor devices (HS code 854140) also rose significantly, from three to ten in the most recent period, with seven linked to China, two to Malaysia, and one to Chinese Taipei. Among importer economies, Türkiye saw the largest increase in total dependencies (from one to seven), while Japan experienced the largest decrease (from seven to three) (Figure 3.16).

Figure 3.17. Number of trade dependencies across segments

Note: Dependencies are calculated based on the methodology explained in Box 1.

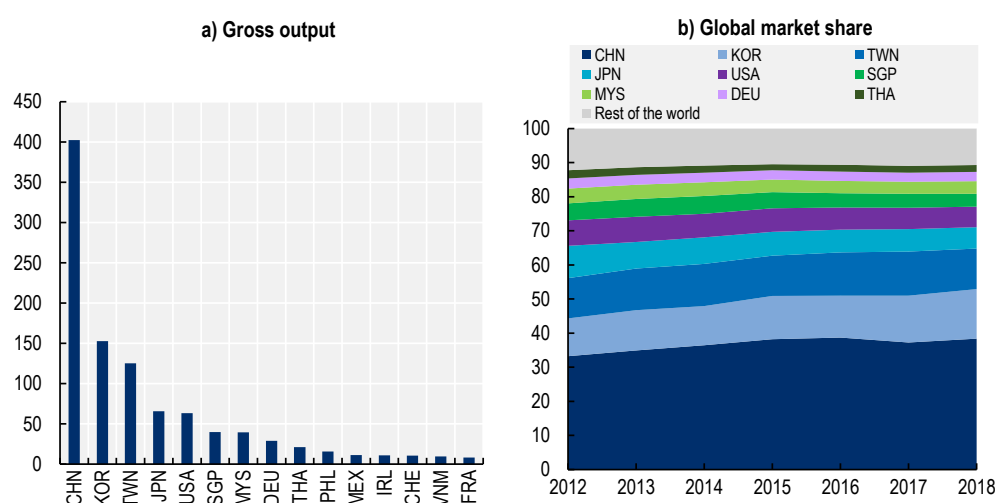
Source: OECD calculations based on UN Comtrade database.

Input-output relationships and links across industries and main economies

This section presents gross output trends and intermediate input flows, highlighting the main connected sectors and economies, based on data from ICIO⁸ and building on the work by Haramboure et al. (2023^[51]). Semiconductor gross output (encompassing design, front-end and back-end manufacturing activities) is highly concentrated in East Asia (Figure 3.18. left panel). China, Japan, Korea and Chinese Taipei account for 71% of global semiconductor gross output, led by China with USD 402 billion in 2018 (38%). These shares remained rather stable from 2012 to 2018.⁹ In comparison, Southeast Asian economies (Singapore, Malaysia, Thailand, the Philippines, and Viet Nam) account for 12% of global gross output (Figure 3.18, right panel).¹⁰

Figure 3.18. Gross output in the semiconductor industry, 2012-2018

Panel a: USD billion; panel b: % of global gross output



Note: The left panel shows the top 15 economies by gross output. The right panel shows the main economies by gross output market share, and the Rest of the World. East Asian economies (China, Japan, Korea, and Chinese Taipei) are shaded in blue, Southeast Asian economies (Singapore, Malaysia and Thailand) in green, and others (Germany and the United States) in purple.

Source: OECD calculations based on ICIO tables and Haramboure et al. (2023^[51]).

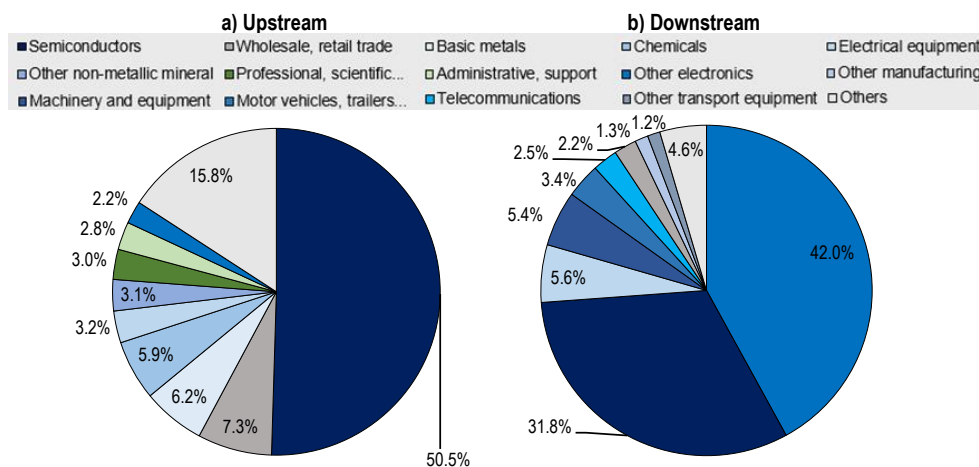
Examining upstream and downstream flows of intermediate materials helps understand the semiconductor industry's reliance on other sectors and economies.¹¹ Upstream connected activities encompass earlier production segments, including sourcing critical components, such as chemicals and basic metals, and the design segment of Integrated Circuits (IC), which defines the product specifications and shapes the subsequent segments of the value chain. Downstream flows, on the other hand, reflect later production segments, such as Assembly, Testing and Packaging (ATP) and their integration into final goods across other industries. The electronics sector, for example, employs the finished chips as an intermediate input to create smartphones, computers and other final products, which are then sold to end costumers.

The semiconductor industry itself encompasses diverse production processes involving both upstream and downstream flows. Over half of the upstream intermediate consumption in the semiconductor industry is sourced internally (Figure 3.19, left panel), reflecting its self-reliance and interdependencies among different segments of the GVC. Other sectors that supply raw materials (e.g., basic metals with 6.2% of upstream intermediate material flows and chemicals with 5.9%) and intellectual services (e.g., professional, scientific, and technical services with 3% and administrative and support services with 2.8%) also play a key role.¹²

Downstream flows reflect the primary sectors using semiconductors. The electronics sector receives 42% of these flows (Figure 3.19, right panel). Almost one third (31.8%) of the flows remain within the semiconductor industry, indicating a high level of internal consumption (e.g. ATP). Other sectoral destinations include manufacturing of electrical equipment (5.6%) and machinery and equipment (5.4%).

Figure 3.19. Upstream and downstream connected sectors, 2018

Panel a: % of total intermediate goods upstream flows; panel b: % of total intermediate goods downstream flows



Note: The left panel shows the sectors with the largest share of upstream flows of intermediate materials. The right panel shows the sectors with the largest share of downstream flows of intermediate materials.

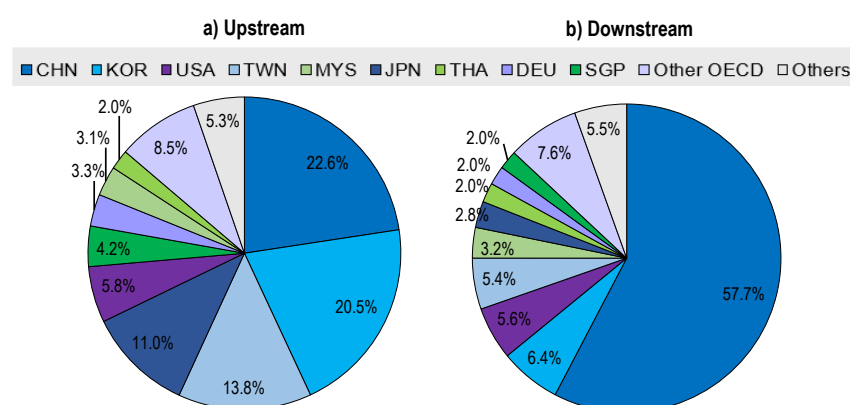
Source: OECD calculations based on ICIO tables and Haramboure et al. (2023^[51]).

East Asia is the primary sourcing region, with China and Korea each accounting for around one fifth of all upstream material flows (Figure 3.20, left panel). Chinese Taipei and Japan also contribute considerably with 14% and 11%, respectively. Downstream flows of intermediate materials are more concentrated, with China representing 58%, almost nine times more than the next largest destination, Korea (6%) (Figure 3.20, right panel).

Further down the value chain, processes like back-end manufacturing become increasingly concentrated in a small number of economies. This concentration emphasises the need of analysing downstream intermediate material flows to identify economies with significant presence in later segments of semiconductor manufacturing processes. As Figure 3.21 shows that over half of the downstream flows towards the same semiconductor industry happen in China and are up to 70% for the other electronics industry. The downstream flows to other semiconductor-using industries are less geographically concentrated, with China receiving 44% and the United States 19%.

Figure 3.20. Main upstream and downstream economies in the semiconductor industry, 2018

Panel a: % of total intermediate goods upstream flows by sourcing economy; panel b: % of total intermediate goods downstream flows by destination economy

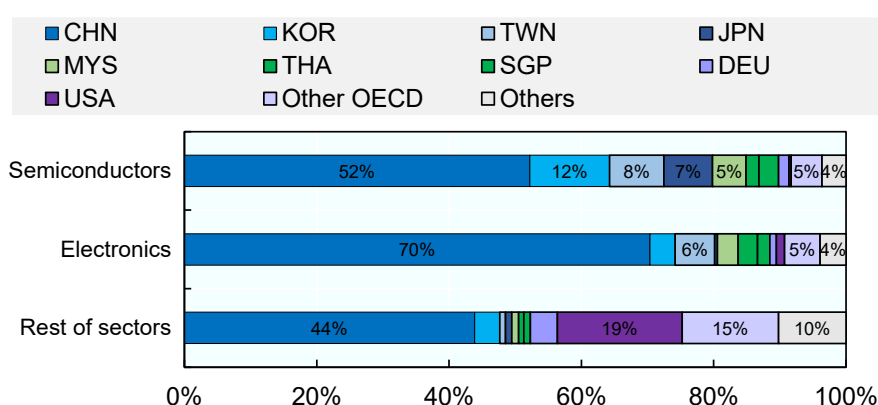


Note: The left panel shows the nine economies by share of upstream flows. The right panel shows the nine economies by share of downstream flows. East Asian economies (China, Japan, Korea and Chinese Taipei) are shared in blue, Southeast Asian economies (Singapore, Malaysia and Thailand) in green, the United States, Germany and other OECD Member countries (other than Germany, Japan, Korea and the United States) in purple and other economies in grey.

Source: OECD calculations based on ICIO tables and Haramboure et al. (2023^[51]).

Figure 3.21. Downstream flows of the semiconductor industry, 2018

% of total downstream transactions from the semiconductor sector



Note: The bars represent the semiconductor industry's downstream flows towards i) the semiconductor industry itself, ii) the electronics industry and iii) all the other industries. Other OECD excludes Germany, Japan, Korea and the United States. East Asian economies (China, Japan, Korea and Chinese Taipei) are shaded in blue, Southeast Asian economies (Singapore, Malaysia and Thailand) in green, the United States, Germany and other OECD Member countries (other than Germany, Japan, Korea and the United States) in purple and other economies in grey.

Source: OECD calculations based on ICIO tables and Haramboure et al. (2023^[51]).

4 Conclusion

This paper examines the structure of the semiconductor value chain and helps identify bottlenecks and vulnerabilities. It offers a detailed analysis of how the different components and processes within the value chain are interconnected, revealing the complex interdependencies that characterise the industry. This report highlights the concentration of critical inputs in particular regions and the specialisation of certain economies in specific inputs.

The analysis shows that the concentration of production and bilateral trade dependencies have been increasing since 2012. The supply of borates is highly concentrated, with Türkiye as a key supplier. Similarly, the supply of silicon wafers and various inputs for wafer fabrication is highly concentrated with Japan and other Asian producers such as Korea and Chinese Taipei playing key roles. Detailed analysis of bilateral trade flows shows that, for numerous products, many economies increasingly rely on China, with limited alternative suppliers. This reliance creates dependencies not only for semiconductors themselves but also for several inputs along the value chain.

The semiconductor industry is uniquely self-reliant: OECD ICIO data show that about half of the intermediate inputs used globally in the semiconductor industry are components sourced from within the industry itself. In parallel, approximately one-third of intermediate outputs are sold to other semiconductor firms for further processing. This dynamic highlights the layered and iterative nature of semiconductor production. In other words, the industry is deeply interconnected: firms rely on one another not just for components, but also for successive processing stages—sending parts back and forth as they are gradually assembled, refined, or integrated into final semiconductor products. Additionally, the level of concentration in downstream intermediate materials flows (particularly to China) highlights vulnerabilities to potential supply chain disruptions.

The concentration of key semiconductor inputs and the growing trade dependencies highlight the need for strategies to diversify supply sources and reduce vulnerabilities. Efforts to diversify supply sources can help decrease trade dependencies, while further enhancing international collaboration and knowledge sharing will be key to address vulnerabilities and manage potential supply chain disruptions.

The findings from these analyses can be complemented by production data analyses, as well as future research focusing on two key areas. First, a more detailed mapping of different segments of the value chain could further enhance understanding of interdependencies. Second, incorporating perspectives from semiconductor-using industries could further refine the mapping, offering a more targeted view of the functioning of the value chain. Together, these efforts would provide deeper insights into both the upstream and downstream elements of the value chain, contributing to a more comprehensive and extensive analysis.

The OECD supports these efforts by analysing different segments of the value chain, by providing policymakers with data to help identify bottlenecks, monitor market imbalances, and manage disruptions in the value chain, and by supporting international dialogue and exchange on best policy practices.

Endnotes

¹ This document uses the terms semiconductor and chips interchangeably.

² The SCAN methodology developed by Bonnet and Ciani (2023^[52]) for a selected group of countries is an example of a more granular analysis.

³ The list of Harmonized System (HS) codes described in this section follows the 2012 HS version, which was in use 2012-17. To compile data for the timeframe of interest (2012-22), the codes from the 2017 and 2022 HS classification versions were mapped to the 2012 HS version using the conversion tables provided by the United Nations (UN), see <https://unstats.un.org/unsd/classifications/Econ>.

⁴ For details on the main results from Haramboure et al. (2023^[51]) and the employed methodology, please refer to Box A C.1.

⁵ The semiconductor industry is not adequately captured by standard classifications, such as ISIC (International Standard Industrial Classification). Building on previous research by Haramboure et al. (2023^[51]), the ISIC division 26 (*manufacture of computer, electronic and optical products*) is divided in two subindustries: i) the semiconductor industry and ii) ICT and other electronics.

⁶ P contains all the goods available in the UN Comtrade database.

⁷ The HS6 classification does not permit differentiation between packaged and unfinished chips. Consequently, identical chips undergoing front-end and back-end manufacturing processes in two distinct economies will be counted twice in gross exports reported in this section.

⁸ Data do not cover investment goods such as machinery and equipment.

⁹ At the time of writing, Haramboure et al. (2023^[51]) used data up to 2018, the most recent year available and for which the technical coefficients remained stable over time. While an update could include data up to 2020, it would capture the disruptive impact of the COVID-19 pandemic and the strategic decisions made to enhance semiconductor GVC resilience.

¹⁰ Value added follows a similar trend to gross production (results are available upon request). Gross production in the semiconductor industry can provide an early sign of final goods movement across the electronics sector, reflecting shifts in upstream activity that can eventually meet final demand (Figure A C.1 in Annex C). See also Annex C for more details on the methodology.

¹¹ As intermediate material flows are derived from technical coefficients following Haramboure et al. (2023^[51]) and splitting the ISIC division 26 (*manufacture of computer, electronic and optical products*) into

i) semiconductors and ii) ICT and other electronics, specific types of inputs, end-products or firms cannot be identified using this dataset. Data on raw materials, inputs for wafer fabrication, silicon wafers, foundry inputs, manufacturing equipment, chips and photosensitive semiconductors are analysed using UN Comtrade.

¹² The sectoral aggregation follows the 2-digit ISIC classification and activities expanding several economic activities, such as *construction of public infrastructure*, or which are more granular (e.g., 3-digit codes, such as *manufacture of medical and dental instruments and supplies*) cannot be represented as sources or destinations.

References

- Antràs, P. et al. (2012), “Measuring the Upstreamness of Production and Trade Flows”, *American Economic Review*, Vol. 102/3, pp. 412-416, <https://doi.org/10.1257/aer.102.3.412>. [69]
- Arjona, R., W. Connell García and C. Herghelegiu (2023), *An enhanced methodology to monitor the EU's strategic dependencies and vulnerabilities*, <https://doi.org/10.2873/768035>. [64]
- Arm Holdings plc (2023), *Annual Report*, <https://investors.arm.com/static-files/dcdd6629-24bb-40ef-ba55-8aca1362205a>. [24]
- Ayodele, A. (2023), *Die Attach: A Comprehensive Guide*, <https://www.wevolver.com/article/die-attach-a-comprehensive-guide>. [50]
- BCG, SIA (2022), *The Growing Challenge of Semiconductor Design Leadership*, https://www.semiconductors.org/wp-content/uploads/2022/11/2022_The-Growing-Challenge-of-Semiconductor-Design-Leadership_FINAL.pdf. [16]
- Bloom, N. et al. (2020), “Are Ideas Getting Harder to Find?”, *American Economic Review*, Vol. 110/4, pp. 1104-1144, <https://doi.org/10.1257/aer.20180338>. [3]
- Bonnet, P. and A. Ciani (2023), *Applying the SCAN methodology to the Semiconductor Value Chain*, <https://publications.jrc.ec.europa.eu/repository/handle/JRC133736>. [52]
- Burkacky, O., T. Kim and I. Yeom (2023), *The growth of advanced semiconductor packaging* | McKinsey, <https://www.mckinsey.com/industries/semiconductors/our-insights/advanced-chip-packaging-how-manufacturers-can-play-to-win> (accessed on 11 October 2024). [43]
- Business, Energy and Industrial Strategy Committee (2022), *The semiconductor industry in the UK*, <https://committees.parliament.uk/publications/31752/documents/178214/default/>. [66]
- Cadence Design Systems (n.d.), *Semiconductor Packaging Materials*, <https://resources.pcb.cadence.com/blog/2023-semiconductor-packaging-materials>. [49]
- Cadence Design Systems, Inc. (2023), *Annual Report*, <https://d18rn0p25nwr6d.cloudfront.net/CIK-0000813672/2ded770d-c7a1-4ca1-8e2a-fa1c129121a8.pdf>. [22]
- Castellano, R. (2023), *Applied Materials: Market Share Decreased 0.3% In Top 10 Sectors It Serves*, <https://seekingalpha.com/article/4611020-applied-materials-market-share-> [37]

- [decreased-0-3-percent-in-top-10-sectors-it-serves](#) (accessed on 11 October 2024).
- Center for Security and Emerging Technology (2021), *The Semiconductor Supply Chain: Assessing National Competitiveness*, Center for Security and Emerging Technology, <https://doi.org/10.51593/20190016>. [10]
- Chen, C. et al. (2020), *Xuantie-910: A Commercial Multi-Core 12-Stage Pipeline Out-of-Order 64-bit High Performance RISC-V Processor with Vector Extension : Industrial Product*, <https://doi.org/10.1109/ISCA45697.2020.00016>. [9]
- Ciani, A. and M. Nardo (2022), *The position of the EU in the semiconductor value chain: evidence on trade, foreign acquisitions, and ownership*, <https://joint-research-centre.ec.europa.eu/system/files/2022-04/JRC129035.pdf>. [63]
- Clarke, P. (2024), *Semiconductor IP market kept growing in 2023*, <https://www.eenewseurope.com/en/semiconductor-ip-market-kept-growing-in-2023/>. [27]
- Congressional Research Service (2023), *Semiconductors and the Semiconductor Industry*, <https://crsreports.congress.gov/product/pdf/r/r47508>. [60]
- Emerging Technology Observatory (2022), *Supply Chain Explorer: Advanced Chips*, <https://chipexplorer.eto.tech/>. [11]
- Esteve, E. (2024), *Semi Market Decreased by 8% in 2023... When Design IP Sales Grew by 6%!*, SemiWiki, <https://semiwiki.com/semiconductor-services/ipnest/343875-semi-market-decreased-by-8-in-2023-when-design-ip-sales-grew-by-6/>. [5]
- European Union (2023), *Regulation (EU) 2023/1781 of the European Parliament and of the Council of 13 September 2023 establishing a framework of measures for strengthening Europe's semiconductor ecosystem and amending Regulation (EU) 2021/694 (Chips Act) (Text with EEA relevance)*, <http://data.europa.eu/eli/reg/2023/1781/oj>. [62]
- Ferguson, J. (2018), *The Process Design Kit: Protecting Design Know-How*, <https://semiengineering.com/the-process-design-kit-protecting-design-know-how/> (accessed on 10 October 2024). [12]
- Fund, T. (2024), *Atomic layer deposition, next-gen transistors, and ASM*, <https://www.techinvestments.io/p/atomic-layer-deposition-next-gen> (accessed on 11 October 2024). [38]
- GlobalFoundries Inc. (2023), *Annual Report*, <https://investors.gf.com/static-files/3b93d28a-1eef-44f9-9c38-d34e8c2b5842>. [20]
- Government Accountability Office (2022), *Semiconductor Supply Chain: Policy Considerations from Selected Experts for Reducing Risks and Mitigating Shortages*. [59]
- Halbleiter (2024), *Semiconductor Technology from A to Z*, <https://www.halbleiter.org/>. [32]
- Haramboure, A. et al. (2023), "Vulnerabilities in the semiconductor supply chain", *OECD Science, Technology and Industry Working Papers*, No. 2023/05, OECD Publishing, Paris, <https://doi.org/10.1787/6bed616f-en>. [51]
- Hars, A. (2022), *Wafer Cleaning Becomes Key Challenge In Manufacturing 3D Structures*, [41]

- <https://semiengineering.com/wafer-cleaning-becomes-key-challenge-in-manufacturing-3d-structures/> (accessed on 11 October 2024).
- Hess, J. and J. Kleinhans (2022), "Governments' role in the global semiconductor value chain #2", <https://www.stiftung-nv.de/en/publication/eca-mapping#4>. [56]
- Hua Hong Semiconductor Limited (2023), *2023 Annual Report*, [21]
https://www.huahonggrace.com/html/ir_reports.php?year=2023.
- Huang, N. (2023), *More than half of the world's SiC wafers might come from China in 2024*, [29]
<https://www.digitimes.com/news/a20231023PD210/silicon-carbide-china-compound-semiconductor-ic-manufacturing-sic.html> (accessed on 10 October 2024).
- Huff, M. (2021), "Recent Advances in Reactive Ion Etching and Applications of High-Aspect-Ratio Microfabrication", *Micromachines*, Vol. 12/8, p. 991, [39]
<https://doi.org/10.3390/mi12080991>.
- IDC (2023), *Worldwide Semiconductor OSAT Market Grew 5.1% YoY in 2022, Growth Expected in 2024 Due to Accumulated Advanced OSAT Demand*, [47]
<https://www.idc.com/getdoc.jsp?containerId=prAP51065123> (accessed on 11 October 2024).
- Kleinhans, J. (2022), "The EDA Chokepoint Dilemma? Openness, Oligopolies, and China's Ecosystem", *IGCC*, https://ucigcc.org/wp-content/uploads/2022/12/2022_wp9_kleinhans-v3-FINAL.pdf (accessed on 10 October 2024). [26]
- Kleinhans, J. (2021), *The lack of semiconductor manufacturing in Europe*, [4]
https://www.interface-eu.org/storage/archive/files/eu-semiconductor-manufacturing.april_2021.pdf (accessed on 12 September 2024).
- Kowalski, P. and C. Legendre (2023), *Raw materials critical for the green transition*, [54]
<https://doi.org/10.1787/c6bb598b-en>.
- Loghin, D. (2024), "Are ARM Cloud Servers Ready for Database Workloads? an Experimental Study", *IEEE Transactions on Cloud Computing*, Vol. 12/3, pp. 818-829, [8]
<https://doi.org/10.1109/TCC.2024.3393895>.
- Ma, W. (2022), *At Apple, a War for Chip Talent Intensifies — The Information*, [15]
<https://www.theinformation.com/articles/inside-apples-war-for-chip-talent> (accessed on 10 October 2024).
- mks (2024), *Silicon Wafer Production*, <https://www.mks.com/n/silicon-wafer-production> [30]
 (accessed on 10 October 2024).
- Mujtaba, H. (2024), *Intel CEO Confirms TSMC's Process Nodes To Power Next-Gen CPUs: N3 For Arrow Lake & N3B For Lunar Lake*, <https://wccfttech.com/intel-ceo-confirms-tsmc-process-nodes-next-gen-cpus-n3-arrow-lake-n3b-lunar-lake/> (accessed on 10 October 2024). [6]
- Nano Institute (2020), *Australian Semiconductor Sector study*, NSW Government Chief Scientist and Engineer, [67]
https://www.chiefscientist.nsw.gov.au/data/assets/pdf_file/0005/339647/Australian-

[Semiconductor-Sector-Study.pdf](#).

- Nenni, D. (2022), *Ansysis' Emergence as a Tier 1 EDA Player— and... - SemiWiki*, [13]
<https://semiwiki.com/eda/ansys-inc/318464-ansys-emergence-as-a-tier-1-eda-player-and-what-that-means-for-3d-ic/> (accessed on 10 October 2024).
- OECD (2024), *Chips, nodes and wafers: A taxonomy for semiconductor data collection*, [1]
 OECD Publishing, Paris, <https://doi.org/10.1787/f68de895-en>.
- OECD (2024), *Inter-Country Input-Output tables*, <https://www.oecd.org/en/data/datasets/inter-country-input-output-tables.html> [68]
 (accessed on 17 October 2024).
- OECD (2023), "Semiconductor informal exchange network: Concept note", *Internal working document*, No. DSTI/CDEP/CIE(2023)1/REV1, OECD, [2]
[https://one.oecd.org/document/DSTI/CDEP/CIE\(2023\)1/REV1/en/pdf](https://one.oecd.org/document/DSTI/CDEP/CIE(2023)1/REV1/en/pdf) (accessed on 4 September 2023).
- OECD (2019), *Measuring distortions in international markets: The semiconductor value chain*, [53]
<https://doi.org/10.1787/8fe4491d-en>.
- Patel, D. (2022), *The Future Of Packaging Gets Blurry*, <https://www.semianalysis.com/p/the-future-of-packaging-gets-blurry> [46]
 (accessed on 11 October 2024).
- Rühlig, T. (ed.) (2024), *Lithography: Is the EU's Semiconductor Manufacturing Equipment a Strategic Chokepoint?*, https://dgap.org/system/files/article_pdfs/DPC%20-%20GESAMT_Final.pdf [33]
 (accessed on 10 October 2024).
- SEMI (2024), *Worldwide Silicon Wafer Shipments and Revenue Fall in 2023*, *SEMI Reports* | [28]
 SEMI, <https://www.semi.org/en/news-media-press-releases/semi-press-releases/worldwide-silicon-wafer-shipments-and-revenue-fall-in-2023-semi-reports>
 (accessed on 10 October 2024).
- Shilov, A. (2023), *The Golden Age of Custom Silicon Draws Near: Part 3 - EE Times*, [14]
<https://www.eetimes.com/the-golden-age-of-custom-silicon-draws-near-part-3/> (accessed on 10 October 2024).
- SIA (2021), *Before the Bureau of Industry and Security, Office of Technology Evaluation, U.S. Department of Commerce, In Response to a Notice by Bureau of Industry and Security, Risks in the Semiconductor Manufacturing and Advanced Packaging Supply Chain*, [35]
<https://www.semiconductors.org/wp-content/uploads/2021/04/4.5.21-SIA-supply-chain-submission.pdf>.
- Siltronic (2024), *Investor presentation*. [31]
- SMIC (2024), *2023 Annual Report*, <https://www.smics.com/uploads/66151b37/e00981.pdf>. [19]
- Stanford Advanced Materials (2023), *Sputtering Target: An Important Component of Semiconductor Materials*. [36]
- Synopsys, Inc (2023), *Annual Report*, [23]
https://s201.q4cdn.com/778493406/files/doc_financials/2023/ar/bafe7900-ebaf-4ae1-8a99-579158b10cb9.pdf.

- The White House (2021), *Building Resilient Supply Chains, Revitalizing American Manufacturing, and Fostering Broad-Based Growth, 100-Day Reviews under*. [57]
- Trade and Technology Council (2022), *U.S.-EU Joint Statement of the Trade and Technology Council, 16 May 2022*, <https://www.whitehouse.gov/wp-content/uploads/2022/05/TTC-US-text-Final-May-14.pdf>. [65]
- TSMC (2024), *Annual Report 2023 (I)*, https://investor.tsmc.com/sites/ir/annual-report/2023/2023%20Annual%20Report_E.pdf. [17]
- Tsutsumi, K. (2024), *Tokyo Electron takes aim at NAND etching leader Lam Research*, <https://asia.nikkei.com/Business/Tech/Semiconductors/Tokyo-Electron-takes-aim-at-NAND-etching-leader-Lam-Research> (accessed on 11 October 2024). [40]
- U.S. Department of Commerce (2023), *Assesment of the status of the microelectronics industrial base in the United States*, <https://www.bis.doc.gov/index.php/documents/technology-evaluation/3402-section-9904-report-final-20231221/file>. [61]
- UMC (2023), *Annual Report*, https://www.umc.com/upload/media/08_Investors/Annual_Reports/Form-20_pdf_chi/Form_20-F_December_31_2023.pdf. [18]
- US Department of Energy (2022), *Semiconductor, Supply Chain Deep Dive Assessment*, <https://www.energy.gov/sites/default/files/2022-02/Semiconductor%20Supply%20Chain%20Report%20-%20Final.pdf>. [58]
- US Geological Survey (2024), *Mineral commodity summaries 2024*, US Geological Survey, <https://doi.org/10.3133/mcs2024>. [34]
- Vardaman, E. (2021), *Too Many Package Options: What Makes Sense for Your Application?*. [44]
- VerWey, J. (2022), *Re-Shoring Advanced Semiconductor Packaging*, Center for Security and Emerging Technology, <https://doi.org/10.51593/20220024>. [45]
- Vicard, V. and P. Wibaux (2023), *EU Strategic Dependencies: A Long View*. [55]
- Wen, T., R. McKay and H. Ma (2022), *Productive Technologies Limited: Company Report*. [42]
- Yole Group (2023), *Advanced semiconductor packaging: A new geopolitical battleground*, <https://www.yolegroup.com/press-release/advanced-semiconductor-packaging-a-new-geopolitical-battleground/> (accessed on 11 October 2024). [48]
- Yole Group (2022), *Power SiC ecosystem is reshaped by vertical integration*, <https://www.yolegroup.com/press-release/power-sic-ecosystem-is-reshaped-by-vertical-integration/> (accessed on 11 September 2024). [7]
- Zeoli, C. (2024), *Synopsys and Cadence: The \$160B Unsung Giants of Semiconductor Design*, <https://www.datagravity.dev/p/synopsys-and-cadence-the-160b-unsung> (accessed on 10 October 2024). [25]

Annex A. Available data and data sources

This subsection focuses on evaluating the existent data sources pertinent to each segment of the value chain, with the objective of identifying any potential gaps in information. For specific details regarding the various databases and related sources, please refer to Table A A.1.

While detailed data at a granular level is available from multiple sources for Table A A.1 upstream and front-end manufacturing, the back-end manufacturing segment is recognised as an area where data availability is comparatively limited. Similarly, for R&D and Design, the coverage is notably constrained. Participants of the network are encouraged to contribute with any additional information that facilitates addressing these information gaps and enhances the comprehensiveness of the available data.

Upstream

- **Materials Market Data**, produced by SEMI. This database provides information related to the revenue and/or shipment data for materials used in front-end and back-end manufacturing processes. It includes 10 segments for wafer-fab related materials and 8 segments for packaging related materials. The dataset contains 10 years of historical data and two years of forecast. It is updated quarterly.
- **Worldwide Semiconductor Equipment Market Statistics**, produced by SEMI. This dataset provides historical data, from 1991 until 2022, of global equipment bookings across 24 equipment categories, broken out by region. It is updated annually.
- **Semiconductor Equipment Database**, produced by TechInsights Market Research Services. This database includes market shares and forecasts for semiconductor equipment markets. The dataset contains annual and quarterly data series.
- **Mass Flow Controller Market Statistics**, produced by SEMI. This dataset contains monthly information on thermal, pressure and liquid Mass Flow Controllers sold to the microelectronics industry. It is updated monthly, delivered six weeks after the close of each month.
- **Inventory on export restrictions on Industrial Raw Materials**, produced by the OECD. It contains information on export regulations in the raw materials sector. It records measures known to restrain export activity from 2009-2021 at the 6-digit level of HS2007 classification. The last update was in 2022.
- **Raw Materials Information System (RIMS)**, produced by the European Commission. The RMIS covers a broad range of thematic areas, including policy and legislation, sectoral and technology demand, supply risk and autonomy, criticality and resilience, circular economy and secondary raw materials, economic and trade-related analyses, material flow analyses, environmental and social sustainability. The information can be accessed by material, sector/technology, and country as well as by thematic area.
- **Directory of Chemical Producers**, produced by IHS Global. This database contains data on chemical producers, plant location, ownership, products, and plant-by-plant capacity information. The database is updated annually. While not explicitly tailored for semiconductor monitoring, it holds valuable information for monitoring critical inputs for semiconductor production.

- **Semiconductor Silicon Demand Forecast Tool**, produced by Omdia. This database contains a forecast of silicon consumption by technology, wafer size, feature size, and market segment. It contains historical data from 2000 and an annual 5-year forecast. The database is updated quarterly.
- **Wafer Demand Database**, produced by Semico Research. This database tracks wafer demand by technology node and semiconductor product type. It contains historical data from 2010 and forecast until 2027. This database is updated quarterly.

R&D and Design

- **Electronic Design Market Data**, produced by SEMI. This data source provides quarterly data related to the revenue and headcount trends within the Electronic Design Automation (EDA) industry. The data encompasses insights into companies involved in providing services, semiconductor intellectual property (IP), and other support functions that facilitate the design of integrated circuits. This database is updated quarterly.
- **PATSTAT**, produced by Directorate for Science, Technology, and Innovation of the OECD. This database is useful to monitor patent filings and approvals related to the semiconductor design. The PATSTAT database contains patent counts by technology. European Patent Office (EPO), US Patent and Trademark Office (USPTO), Patent Co-operation Treaty (PCT) and Patent Families indicators are presented according to classes of the International Patent Classification. This database is updated twice a year and the last update was in October 2023.
- **Crunchbase**. Crunchbase is a platform for companies, startups, and associated individuals. This database includes extensive coverage of startups that participate in the R&D step of the semiconductor value chain. It contains information on their industry, location, founding date, and contact details. Additionally, it contains extensive information on risk financing, including valuations and investor type.
- **Dealroom**. Dealroom is a platform that provides data related to startups, scale-ups, and venture capital activity. It provides information on the company, the industry, founding data, the founder, and the background of the founder. Thereby, this database is useful for identifying innovation activities in the R&D and Design segment and analysing the associated characteristics.

Front-end manufacturing

- **World Fab Watch**, produced by SEMI. This database tracks more than 1 450 facilities worldwide and provides a snapshot of the last quarter of all front-end semiconductor fabs. It contains useful fab-specific details such as fab type, start year, equipment spending, capacity per month or wafer size production.
- **200mm Fab Outlook**, produced by SEMI. This dataset tracks operations and investments in 200mm semiconductor fabs. It tracks more than 330 fabs by over 150 companies and includes quarterly historical data and forecast until 2026.
- **300mm Fab Outlook**, produced by SEMI. This dataset tracks operations and investments in 300mm semiconductor fabs. It tracks more than 387 fabs/lines worldwide by more than 82 companies and includes quarterly historical data and forecast until 2026.
- **World Fab Forecast**, produced by SEMI. This dataset tracks worldwide fab spending, construction, production capacity, and technology trends by device type. It tracks more than 1 450 facilities worldwide owned/operated by more than 550 companies. The technologies covered

include semiconductors, discretes, optoelectronics, LEDs, MEMS, and sensors. It includes fabs that use both 200mm and 300mm wafer sizes. Moreover, it includes quarterly forecast data on fab investment, capacity, and other related aspects.

- **Semico Fab Database**, produced by Semico Research. This database tracks operations and investments in semiconductor fabs. It covers fabs that are planned, under construction, installing tools, operating, closing, and closed. Semico issues an updated report annually.
- **Pure-play Foundry Market Tracker**, produced by Omdia. This database provides information on Pure-Play foundries' business performance, manufacturing capacity, and industry forecasts. It is updated quarterly.

Back-end manufacturing (Assembly, Testing and Packaging)

- **Worldwide Assembly & Test Facility Database**, produced by SEMI. This database covers back-end facilities of IDMs and OSATs, with packaging technology offerings by manufacturing location. It is updated annually.

Table A A.1. Detail on data sources

Database	Source	Link	Segment
Materials Market Data	SEMI	SEMI® Semiconductor Materials Market Information SEMI	Upstream
Worldwide Semiconductor Equipment Market Statistics	SEMI	World Fab Watch SEMI	
Semiconductor Equipment Database	TechInsights	https://www.techinsights.com/market-analysis/chip-market-research#equipment-database	
Mass Flow Controller Market Statistics	SEMI	Mass Flow Controller Market Statistics SEMI	
Inventory on export restrictions on Industrial Raw Materials	OECD	Export restrictions on Industrial Raw Materials (oecd.org)	
Raw Materials Information System (RIMS)	European Commission	https://rmis.jrc.ec.europa.eu/	
Directory of Chemical Producers	S&P Global	Directory of Chemical Producers S&P Global (spglobal.com)	
Silicon Demand Forecast Tool	Omdia	https://omdia.tech.informa.com/collections/silicon-demand-forecast-tool	
Wafer Demand	Semico Research	Semico Wafer Demand: Q2 2023 Highlights Semico Research	
Silicon Wafer Market Monitor	SEMI	Silicon Wafer Market Monitor SEMI	
Photomask Market Characterization Study	SEMI	Photomask Market Characterization Study SEMI	
National Minerals Information Center	U.S Geological Survey	National Minerals Information Center U.S. Geological Survey (usgs.gov)	
Critical Minerals Demand Dataset	IEA	Critical Minerals Demand Dataset - Data product - IEA	
Electronic Design Market Data	SEMI	Electronic Design Market Data SEMI	R&D & Design
Crunchbase	Crunchbase	https://www.crunchbase.com/	
PATSTAT	OECD	https://stats.oecd.org/Index.aspx?DataSetCode=PATS_IPC	
Pitchbook	Pitchbook	PitchBook's Data: Private & Public Market Data PitchBook	
World Fab Forecast	SEMI	World Fab Forecast SEMI	
300mm Fab Outlook	SEMI	300mm Fab Outlook to 2026 SEMI	

200mm Fab Outlook	SEMI	200mm Fab Outlook to 2026 SEMI	Front-end manufacturing
World Fab Watch	SEMI	World Fab Watch SEMI	
Pure Play Foundry Market Tracker	Omdia	Pure Play Foundry Market Tracker Omdia (informa.com)	
Global Semiconductor Manufacturing Market Tracker	Omdia	Global Semiconductor Manufacturing Market Tracker Omdia (informa.com)	
Semico Fab Database	Semico Research	Semico Fab Database: Update Summary 2023 Semico Research	
Global Wafer Capacity	Knometaresearch	Knometaresearch Reports	
McClean Report	TechInsights	The McClean Report TechInsights	
Semiconductor Manufacturing Economics	TechInsights	Semiconductor Manufacturing Economics TechInsights	Back-end manufacturing
Worldwide Assembly & Test Facility Database	SEMI	Worldwide Assembly & Test Facility Database, 2022 Edition SEMI	
Semiconductor Manufacturing Economics	TechInsights	Semiconductor Manufacturing Economics TechInsights	

Source: OECD compilation

Annex B. Efforts to map the semiconductor value chain

This subsection provides an overview of recent and ongoing efforts by governments and public sector organisations to map and monitor the semiconductor value chain. This overview is by no means exhaustive and is currently limited to publicly available and more widely publicised initiatives (Hess and Kleinhans, 2022^[56]). Efforts outlined below have differing objectives, coverage, and frequency. They generally have a focus on the respective economy but include both one-time reports and briefings to ongoing monitoring efforts.

United States

In 2021, the White House published its 100-Day Reviews on ‘Building Resilient Supply Chains, Revitalizing American Manufacturing, and Fostering Broad-Based Growth’ (The White House, 2021^[57]). One of the supply chains identified was ‘Semiconductor Manufacturing and Advanced Packaging’ for which the Department of Commerce led the review.

This review starts by mapping the value chain across five segments: “(1) design; (2) fabrication; (3) assembly, test, and packaging (ATP) and advanced packaging; (4) materials; and (5) manufacturing equipment”. It takes a US centric view to understand the performance of the national industry within each of these segments.

The review identifies eight risks across the semiconductor value chain: “(1) fragile supply chains; (2) malicious supply chain disruptions; (3) use of obsolete and generations-old semiconductors and related challenges for continued profitability of companies in the supply chain; (4) customer concentration and geopolitical factors; (5) electronics production network effects; (6) human capital gaps; (7) intellectual property (IP) theft; and (8) challenges in capturing the benefits of innovation and aligning private and public interests.” On that basis, the review offers a number of policy recommendations centred around promoting investment in, collaboration with, and the strengthening of the industry at a national (and international) level, as well as international engagement towards supply chain resilience.

The US Department of Energy published a similar report in 2022 (US Department of Energy, 2022^[58]), although with an additional focus on semiconductors for power electronics. Their findings match those of the Department of Commerce, whose review is referred to on multiple occasions, and find similar patterns and vulnerabilities in the power electronics semiconductors.

In July 2022, the US Government Accountability Office (GAO) published a report titled “Semiconductor Supply Chain: Policy Considerations from Selected Experts for Reducing Risks and Mitigating Shortages” (Government Accountability Office, 2022^[59]), following increasing interest in semiconductor policymaking from both the US Senate and the House of Representatives. The report is less focused on mapping the supply chain and rather looks to identify key risks and relevant policy levers through research and expert interviews.

The recommendations from this report fall under five categories, with one dedicated to supply chain strengthening. The key themes in this section are supply chain security, supply chain monitoring and transparency, secure access to critical materials, building supply chain resiliency, prioritising chip supply based on industry needs, and increasing and clarifying the demand signal.

Building supply chain resilience was identified as one of the most important areas. The suggested levers include mandated resilience standards, targeting of key risks, as well as promoting agility. Among the various recommendations for value chain and supply chain mapping, some experts interviewed suggested using "high-level organisational information" rather than requiring the disclosure of proprietary information from private entities. Others included forming a team dedicated to understanding and monitoring the supply chain.

In 2023, the Congressional Research Service published a report on semiconductors and the semiconductor industry (Congressional Research Service, 2023^[60]) which provides a broader briefing on semiconductors and highlights the global nature of their production. The report also highlights the state of the US industry across the different segments of the value chain as well as planned investments.

In 2023, the United States Department of Commerce published a report assessing the status of the microelectronics industrial base in the United States (U.S. Department of Commerce, 2023^[61]). This report highlights the United States microelectronics industry's leadership but also its challenges. Key areas in the semiconductor value chain such as assembly and packing, along with materials supply, heavily depend on foreign sources. Workforce shortages, foreign competition, and high manufacturing costs are major obstacles, despite strong intellectual property protection and research capabilities.

Among the recent US policy mechanisms implemented is the Semiconductor Alert Mechanism (SAM)¹ which gives stakeholders the opportunity to signal disruptions to the semiconductor value chain.

European Union

The third pillar of the European Chips Act (European Union, 2023^[62]) seeks to establish a mechanism for EU member states to coordinate on semiconductor value chain monitoring, as well as anticipating demand and supply shocks.

In 2022, the European Commission Joint Research Centre (JRC) published a technical report titled: 'The position of the EU in the semiconductor value chain: evidence on trade, foreign acquisitions, and ownership' (Ciani and Nardo, 2022^[63]). The report maps the global semiconductor value chain and looks at the EU's position within it.

The analysis is split into four main segments, describing:

- The EU's position in terms of its trade balance using product level data;
- The global linkages between semiconductor value chain actors using firm-level supply chain data;
- The investment and Mergers and Acquisitions (M&A) activity involving firms in the EU using firm-level data;
- The role of EU-owned firms in each segment of the supply chain using turnover data.

The report finds that the value chain is dominated by a few economies, but that given the complexity and high degree of specialisation, no single actor has dominant position over the entire chain. Therefore, no economy is truly independent when considering the entire value chain. The majority of insights focus on the EU's position, finding a significant role for its companies in most segments of the value chain, with the exception of semiconductor fabrication.

In 2023, the JRC published a paper outlining how it applied the Supply Chain Alert Notification (SCAN) methodology (Bonnet and Ciani, 2023^[52]) proposed by the office of the Chief Economist of the European Commission's DG GROW to the Semiconductor value chain. To identify supply chain risks, the methodology uses a number of structural indicators such as concentration of imports and possible substitution, and high-frequency indicators to capture fluctuations in import prices and quantities. By monitoring these factors, the goal is to identify potential structural vulnerabilities and track imbalances in near-real time.

In the case of semiconductors, the JRC found that among the 74 semiconductor value chain relevant products they examined, ten were considered to be at medium to high-risk of import disruption in 2021. This was primarily driven by factors such as high concentration and low substitutability. According to SCAN, six of these products are of particular concern for the EU in the last quarter of 2022 due to increasing prices and decreasing quantities. These are "i) Azidies, Silicides; ii) Photographic plates and film in the flat for monochrome photography; iii) Sheets and plates of polarising material; iv) Fans of a kind used principally for cooling microprocessors; v) Diodes; vi) Electronic integrated circuits as dynamic random access memories." The JRC's paper then attempts to evaluate the importance of these products to their respective value chain segments.

The JRC's paper assesses the SCAN methodology in its application to the semiconductor value chain. It notes that while it can identify structural vulnerabilities there are some confounding factors. The product level view does not allow to isolate the semiconductor value chain specifically. This is particularly important in the case of raw materials which play an important role in other value chains. Additionally, inventory is not accounted for in the methodology and can lead to delays in the impact of disruptions in the monitored data.

Building on the JRC's work, the DG GROW Chief Economist unit has proposed a new methodology for strategic dependency monitoring (Arjona, Connell García and Herghelegiu, 2023^[64]), which can identify the potential for diversification for specific products, and those particular cases in which the EU needs to develop internal capacity in order to diversify.

Finally, the EU has also set up a warning mechanism for firms experiencing supply chain disruptions called the Semiconductor Alert System.²

US-EU Trade and Technology Council

The May 2022 Trade and Technology Council (TTC) Statement on Semiconductors outlined a common understanding between the United States and the European Union (Trade and Technology Council, 2022^[65]). Both parties recognised the importance given to securing the supply chain by their respective Chips Acts. While the 2021 supply chain disruptions in semiconductors can be traced back to an exceptional confluence of factors, it underlines the importance of information sharing and monitoring.

While the EU and US initiatives outlined above mostly use trade data, in its statement the TTC emphasises the important role that industry can play in monitoring efforts. A commitment is made to working with industry on transparency regarding semiconductor demand to better inform public initiatives. The sensitivity of business confidential information is acknowledged. The parties propose the implementation of a common "early warning and monitoring mechanism", on which further information is not yet available.

Other jurisdictions

Many jurisdictions have commissioned analyses of their respective semiconductor industries to assess capacities and capabilities. Such reports tend to acknowledge the complexity of the global value chains involved but do not dedicate significant sections to mapping and monitoring efforts.

In the United Kingdom (UK), the UK's Business, Energy and Industrial Strategy Committee published a report on 'The semiconductor industry in the UK' in November 2022 (Business, Energy and Industrial Strategy Committee, 2022^[66]). This report describes the extent of the semiconductor industry in the United Kingdom, where it has strengths and where it is lagging, and its vulnerabilities as part of the global value chain. The main strengths outlined are the UK's design, intellectual property, and advanced material chips. On the other hand, it does not have end-to-end production capabilities and is dependent on concentrated suppliers for many inputs and fabrication. The report also issues a set of policy recommendations for the development of semiconductors around focusing on existing strengths, facilitating the development of semiconductor manufacturing in the country, and bridging the gap with the EU and the United States in terms of monitoring and mitigation efforts.

In 2020, the University of Sydney Nano Institute published a review of the Australian Semiconductor Industry on behalf of the Office of the New South Wales Chief Scientist & Engineer (Nano Institute, 2020^[67]). The goal of this report is to assess the Australian semiconductor industry, its scale, the reasons for its small size, the potential advantages of Australia and, accordingly, the levers that might be used to develop it. One of the areas of improvement suggested by this paper is the development of the semiconductor industry through the value chain lens as a way of promoting industry success in the long term, along with improvements in skill, capital access, research, and market information.

The Network team would welcome documentation from all members about their respective efforts to map and monitor the semiconductor value chain and/or value chain. This would contribute to the shared knowledge of the network and help all participants improve their own initiatives.

Annex C. Inter-Country Input-Output (ICIO) tables

Inter-Country Input-Output (ICIO) tables are the statistical infrastructure developed by the OECD to increase understanding of Global Value Chains (GVCs), by mapping flows of production, consumption, investment and flows of international trade, broken down by economic activity and economy. ICIO tables reflect the division of labour between economies and industries in the production of goods and services and allow for insights into major current dynamics underlying regional and global value chains, such as GVCs resilience (especially after the COVID-19 pandemic), greenhouse gas emissions embodied in internationally traded goods, and trade dependencies (OECD, 2024^[68]).

Nonetheless, for analysing the semiconductor global value chain, the ICIO tables are too aggregated, therefore, an additional step is required to obtain more granular insights. Haramboure et al. (2023^[51]) developed the semiconductor-augmented ICIO tables, which are based on relevant information from five domestic input-output tables (Korea, Japan, Malaysia, Chinese Taipei and the United States) and from bilateral trade data (Bilateral Trade in Goods by Industry and End-use Category) that are sufficiently granular to identify semiconductors (Harmonised Systems headings 8541 and 8542).

To split the ISIC division 26 (*manufacture of computer, electronic and optical products*) into two subindustries (the semiconductor industry and ICT and other electronics), technical coefficients are estimated to determine the proportion of each subindustry in domestic and foreign uses for each producing and purchasing economy. The set of technical input-output coefficients provides enough granularity to analyse the semiconductor industry and focuses on sectoral output and value-added contributions from design, front-end, and back-end manufacturing rather than tracking the intricate details of all upstream inputs such as manufacturing equipment sourcing.

The five economies listed above represent around 75% of the global semiconductor value added in 2018 and their domestic technical coefficients are directly available. For the other economies, technical coefficients are imputed from an economy with a similar semiconductor industry according to expert judgement. One caveat of this analysis is that gross-fixed capital formation is not considered under this approach. Moreover, the study period only extends up to 2018 and therefore does not reflect the disruptive impact of the COVID-19 pandemic or the geopolitical strategic decisions made to increase the resilience of the semiconductor GVCs. Nonetheless, technical coefficients remained rather stable over time, reflecting consistent conditions and highlight the usefulness of the results, even in light of later disruptions. For details on the main results from Haramboure et al. (2023^[51]), please refer to Box A C.1 and for the matrix construction of the semiconductor-augmented ICIO tables and expert judgement criteria for economies-matching, please refer to Haramboure et al. (2023^[51]).

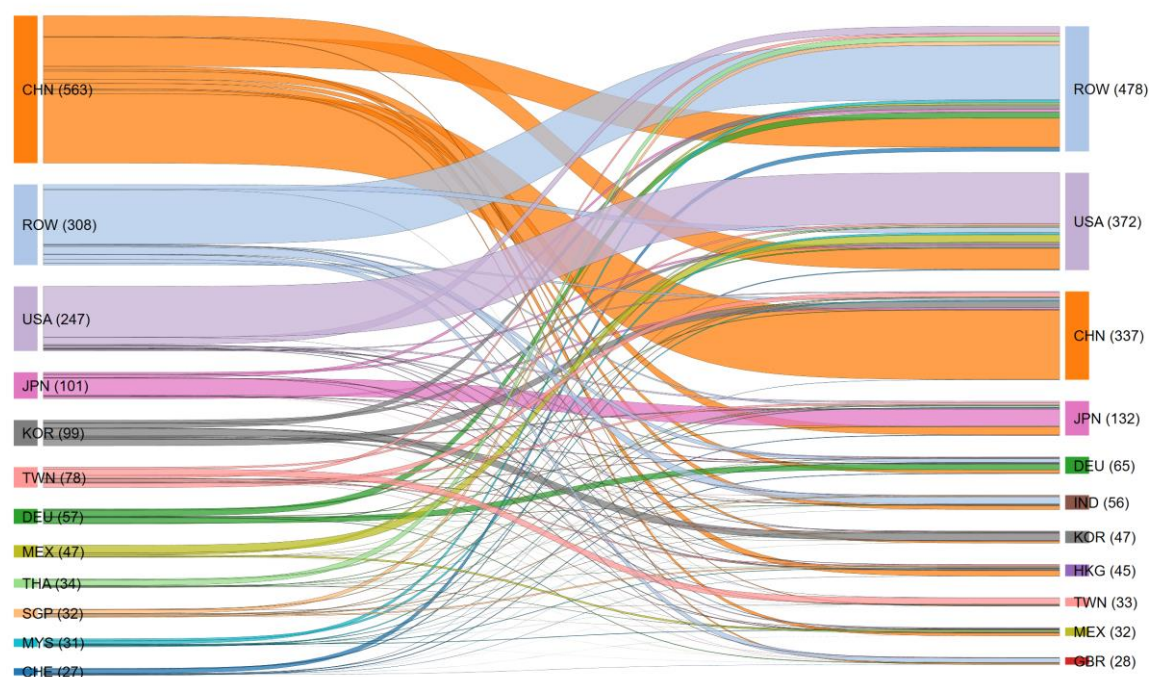
Final goods

Final goods flow in the semiconductor industry are calculated as part of the broader electronics sector, considering the integral role of semiconductors as an input for final electronic goods. While the final good flows of semiconductors might be distributed across multiple sectors beyond electronics, such as automotive and telecommunications, this analysis focuses solely on the primary downstream destination of semiconductor products.

China is the leading source of final electronic goods, with USD 563 billion and serving diverse destinations worldwide in 2018 (Figure A C.1). Additionally, China has strong domestic consumption, making it the second-largest destination in this segment. The United States, in comparison, is the main destination for final electronic goods, driven by robust domestic consumption and diverse sources around the world.

Figure A C.1. Final goods flow in the electronics sector, 2018

USD billion



Note: Semiconductors are not considered as a final good in Haramboure et al. (2023^[51]). For this reason, aggregates come from the broader electronics sector (ISIC Division 26).

Source: OECD calculations based on ICIO tables and Haramboure et al. (2023^[51]).

Box A C.1. Vulnerabilities in the semiconductor value chain

Haramboure et al. (2023^[51]) used the Inter-Country Input-Output (ICIO) data to reveal structural features of the semiconductor value chain, identifying three main takeaways. First, the semiconductor industry is among the most upstream industries, as measured by its distance to final demand (Antràs et al., 2012^[69]). The “ICT and electronics (excluding semiconductors)” sector is highly dependent on semiconductors, with semiconductor value added accounting for 8% of its final demand (compared to 2% for primary energy). Other industries (including motor vehicles) demand semiconductors to a similar degree as primary energy. Second, the semiconductor industry is geographically highly concentrated, with the top five semiconductor-producing economies, mainly in East Asia, contributing around three-quarters of global value added. Lastly, across economies, dependencies on semiconductors are driven by different industries. In most Asian economies, they are explained by “ICT and electronics (excluding semiconductors)”, while in the United States and most of Europe, dependencies reflect to a much larger extent the “Transport equipment” sector and other industries.

Source: Haramboure et al. (2023^[51]).

Annex D. List of HS codes

Table A D.1. List of HS codes associated to raw materials

HS code	Label
252800	Natural borates and concentrates thereof (whether or not calcined), but not including borates separated from natural brine; natural boric acid containing not more than 85 % of H ₃ BO ₃ calculated on the dry weight
280421	Argon
280429	Rare gases (excl. Argon)
280461	Silicon; containing by weight not less than 99.99% of silicon
280480	Arsenic
281000	Oxides of boron; boric acids
281212	Phosphorus oxychloride
282560	Germanium oxides and zirconium dioxide
283325	Sulphates, copper
284011	Borates; disodium tetraborate (refined borax), anhydrous
284019	Borates; disodium tetraborate (refined borax), other than anhydrous
284920	Silicon Carbide
285000	Hydrides, nitrides, azides, silicides and borides, whether or not chemically defined (excl. compounds which are also carbides of heading 2849, and inorganic or organic compounds of mercury whether or not chemically defined)
811292	Gallium, germanium, hafnium, indium, niobium (columbium), rhenium and vanadium; articles thereof, unwrought, including waste and scrap, powders
811299	Articles of niobium "columbium", gallium, indium, vanadium and germanium, n.e.s.

Note: The classifications presented in this table are based on the HS 2017 edition. Subsequent revisions introduced in the HS 2022 edition may have modified or restructured some of the codes listed.

Source: OECD compilation based on Bonnet and Ciani (2023^[52]), Haramboure et al. (2023^[51]) and OECD (2019^[53]).

Table A D.2. List of HS codes associated to inputs for wafer fabrication

HS code	Label
370130	Photographic plates and film in the flat, sensitised, unexposed, with any side > 255 mm
370199	Photographic plates and film in the flat for monochrome photography, sensitised, unexposed, of any material other than paper, paperboard or textiles (excl. X-ray film and photographic plates, film in the flat with any side > 255 mm, and instant print film)
370790	Photographic goods

Note: The classifications presented in this table are based on the HS 2017 edition. Subsequent revisions introduced in the HS 2022 edition may have modified or restructured some of the codes listed.

Source: OECD compilation based on Bonnet and Ciani (2023^[52]), Haramboure et al. (2023^[51]) and OECD (2019^[53]).

Table A D.3. List of HS codes associated to silicon wafers

HS code	Label
381800	Silicon wafers

Note: The classifications presented in this table are based on the HS 2017 edition. Subsequent revisions introduced in the HS 2022 edition may have modified or restructured some of the codes listed.

Source: OECD compilation based on Bonnet and Ciani (2023^[52]), Haramboure et al. (2023^[51]) and OECD (2019^[53]).

Table A D.4. List of HS codes associated to foundry inputs

HS code	Label
900120	Sheets and plates of polarising material/sheets of semiconductor
900190	Lenses, prisms, mirrors and other optical elements, unmounted
900219	Objective lenses
900220	Optical filters
900290	Lenses, prisms, mirrors and other optical elements, mounted
901210	Electron microscopes, proton microscopes and diffraction apparatus
901290	Parts and accessories for electron microscopes, proton microscopes and diffraction apparatus
903141	Optical instruments inspecting semiconductor devices

Note: The classifications presented in this table are based on the HS 2017 edition. Subsequent revisions introduced in the HS 2022 edition may have modified or restructured some of the codes listed.

Source: OECD compilation based on Bonnet and Ciani (2023^[52]), Haramboure et al. (2023^[51]) and OECD (2019^[53]).

Table A D.5. List of HS codes associated to manufacturing equipment

HS code	Label
841459	Air or vacuum pumps (excl. gas compound elevators and pneumatic elevators and conveyors); air or other gas compressors and fans; ventilating or recycling hoods incorporating a fan, whether or not fitted with filters; gas-tight biological safety cabinets, whether or not fitted with filters; parts thereof.
841950	Heat-exchange units (excl. those used with boilers)
842129	Machinery and apparatus for filtering or purifying liquids (excl. such machinery and apparatus for water and other beverages, oil or petrol-filters for internal combustion engines and artificial kidneys)
842139	Machinery and apparatus for filtering or purifying gases (excl. isotope separators and intake air filters for internal combustion engines, and catalytic converters and particulate filters for purifying or filtering exhaust gases from internal combustion engines)
842199	Parts of machinery and apparatus for filtering or purifying liquids or gases, n.e.s.
848610	Machines and apparatus of a kind used solely or principally for the manufacture of semiconductor boules or wafers
848620	Machines and apparatus of a kind used solely or principally for the manufacture of semiconductor devices or of electronic integrated circuits
848630	Machines and apparatus of a kind used solely or principally for the manufacture of flat panel displays
848640	Machines and apparatus of a kind used solely or principally for the manufacture or repair of masks and reticles, assembling semiconductor devices or electronic integrated circuits, or for lifting, handling, loading or unloading items of heading 8486
848690	Machines and apparatus of heading 8486; parts and accessories
900699	Photographic flashlight apparatus
903082	Instruments and apparatus for measuring or checking semiconductor wafers or devices, incl. integrated circuits
903084	Instruments and apparatus for measuring or checking electrical quantities, with recording device (excl. appliances specially designed for telecommunications, multimeters, oscilloscopes and
903300	Regulating or controlling instruments

Note: The classifications presented in this table are based on the HS 2017 edition. Subsequent revisions introduced in the HS 2022 edition may have modified or restructured some of the codes listed.

Source: OECD compilation based on Bonnet and Ciani (2023^[52]), Haramboure et al. (2023^[51]) and OECD (2019^[53]).

Table A D.6. List of HS codes associated to final products (chips)

HS code	Label
854110	Electrical apparatus; diodes, other than photosensitive or light emitting diodes
854121	Electrical apparatus; transistors, (other than photosensitive), with a dissipation rate of less than 1W
854129	Electrical apparatus; transistors, (other than photosensitive), with a dissipation rate of 1W or more
854130	Electrical apparatus; thyristors, diacs and triacs, other than photosensitive devices
854160	Crystals; mounted piezo-electric
854190	Electrical apparatus; parts for diodes, transistors and similar semiconductor devices and photosensitive semiconductor devices
854231	Electronic integrated circuits; processors and controllers, whether or not combined with memories, converters, logic circuits, amplifiers, clock and timing circuits, or other circuits
854232	Electronic integrated circuits; memories
854233	Electronic integrated circuits; amplifiers
854239	Electronic integrated circuits; n.e.c. in heading no. 8542
854290	Parts of electronic integrated circuits
852351	Solid-state, non-volatile data storage devices for recording data from an external source [flash memory cards or flash electronic storage cards]
852352	Cards incorporating one or more electronic integrated circuits "smart cards"
852359	Semiconductor media, unrecorded, for the recording of sound or of other phenomena.
853290	Parts of electrical "pre-set" capacitors, fixed, variable or adjustable, n.e.s
853390	Parts of electrical resistors

Note: The classifications presented in this table are based on the HS 2017 edition. Subsequent revisions introduced in the HS 2022 edition may have modified or restructured some of the codes listed.

Source: OECD compilation based on Bonnet and Ciani (2023^[52]), Haramboure et al. (2023^[51]) and OECD (2019^[53]).

Table A D.7. List of HS codes associated to photosensitive semiconductor devices

HS code	Label
854140	Electrical apparatus; photosensitive, including photovoltaic cells, whether or not assembled in modules or made up into panels, light emitting diodes
854150	Electrical apparatus; photosensitive semiconductor devices n.e.s. in heading no. 8541, including photovoltaic cells, whether or not assembled in modules or made up into panels

Note: Data for 2022 is not available for the HS code 854150 given that this product was declassified in the 2022 update of the HS classification and the conversion tables do not provide a clear mapping with newly classified products. The classifications presented in this table are based on the HS 2017 edition. Subsequent revisions introduced in the HS 2022 edition may have modified or restructured some of the codes listed.

Source: OECD compilation based on Bonnet and Ciani (2023^[52]), Haramboure et al. (2023^[51]) and OECD (2019^[53]).

Endnotes

¹ Microelectronics Early Alert System: <https://www.trade.gov/microelectronics-early-alert-system>

² Semiconductor Alert System: https://ec.europa.eu/eusurvey/runner/Semiconductor_Alert_System